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Removing independent noise in systems neuroscience data using DeepInterpolation

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Progress in many scientific disciplines is hindered by the presence of independent noise. Technologies for measuring neural activity (calcium imaging, extracellular electrophysiology and functional magnetic resonance imaging (fMRI)) operate in domains in which independent noise (shot noise and/or thermal noise) can overwhelm physiological signals. Here, we introduce DeepInterpolation, a general-purpose denoising algorithm that trains a spatiotemporal nonlinear interpolation model using only raw noisy samples. Applying DeepInterpolation to two-photon calcium imaging data yielded up to six times more neuronal segments than those computed from raw data with a 15-fold increase in the single-pixel signal-to-noise ratio (SNR), uncovering single-trial network dynamics that were previously obscured by noise. Extracellular electrophysiology recordings processed with DeepInterpolation yielded 25% more high-quality spiking units than those computed from raw data, while DeepInterpolation produced a 1.6-fold increase in the SNR of individual voxels in fMRI datasets. Denoising was attained without sacrificing spatial or temporal resolution and without access to ground truth training data. We anticipate that DeepInterpolation will provide similar benefits in other domains in which independent noise contaminates spatiotemporally structured datasets.

Independent noise impacts experimental systems neuroscience, given the weakness of neuronal signals relative to the shot and thermal noise present in devices used for observing the brain. For example, in vivo imaging of fluorescent indicators (voltage and calcium) operates in photon-count regimes where shot noise dominates pixel-wise measurements. Similarly, thermal and shot noise present in electronic circuits can impact the detection of action potentials in in vivo electrophysiological recordings and blood-oxygen-level-dependent (BOLD) responses in fMRI, impairing the measurement of authentic biological signals.

When data points in spatiotemporal proximity harbor the same underlying signal but are independently affected by noise, median or Gaussian filters (in the time or Fourier domain) can be used to enhance single-trial dynamics at the expense of reduced spatial and/or temporal resolution. Although this filtering approach is widely used, optimal denoising filters can be complicated to design by hand when data relationships span multiple dimensions (for example, time and space) or are intrinsically nonlinear.

Instead, it may be more efficient to remove noise by learning a model of the complex hierarchical relationships between data points, using a large training dataset[1]. This approach would exploit statistical relationships between samples to reconstruct a noiseless version of the signal, rather than applying filters to remove noise. This approach has been successful for learning denoising or upsampling filters[2,3] but, until recently, was limited to cases for which ground truth data are available. The Noise2Noise[4] approach demonstrated that deep neural networks can be trained to perform image restoration without access to ground truth data, with performance comparable to that of training using cleaned data. Indeed, gradients derived from a loss function calculated on noisy samples are partially aligned toward the correct solution. Provided that the noisy loss is an unbiased estimate of the true loss, the correct denoising filters are learned. However, unlike the data used to illustrate Noise2Noise[4], neuronal data do not consist of pairs of samples with identical signals but different noise.

To overcome this limitation, we adopted an approach similar to the Noise2Self[5] and Noise2Void[6] frameworks. In the absence of training pairs, we solved an interpolation problem to learn spatiotemporal relationships present in the data. We trained a model to learn the underlying relationship between each data point and its neighbors by optimizing the reconstruction loss calculated on each noisy instance of the sample itself. This approach is valid as long as the noise present in the target sample is independent from the input (adjacent) samples; otherwise, our relationship model would overfit the noise that we seek to remove. We eliminated overfitting by omitting the target center frame from the input and by presenting training samples only once during training. During inference, the dataset is streamed through the trained network to reconstruct a nearly noiseless version of the underlying signal.

We applied these principles to the problem of denoising dynamical signals at the heart of systems neuroscience, thereby building a general-purpose denoising algorithm that we call DeepInterpolation. We describe and demonstrate the use of DeepInterpolation for two-photon in vivo calcium (Ca^{2+}) imaging, in vivo electrophysiology and whole brain fMRI. We anticipate that our approach will be applicable to other experimental fields with minimal modifications.

Results

Application of DeepInterpolation to two-photon calcium imaging. In our recordings of the mouse visual cortex, we first confirmed that our two-photon recordings with GCaMP6f, a genetically encoded calcium indicator[7], were dominated by Poisson noise as shown by individual plots of all pixel variances against their means (Supplementary Fig. 1). Next, we constructed our denoising network with a UNet-inspired architecture[8,9] following two simple principles: first, a single pixel can share information with pixels in a local region of the image (our neuronal network architecture had a final receptive field diameter of 78 pixels or 60 µm); and second, the decay dynamics of calcium-dependent fluorescent indicators (GCaMP6f $\tau_{peak} = 80 \pm 35$ ms (s.e.m.), $\tau_{1/2} = 400 \pm 41$ ms

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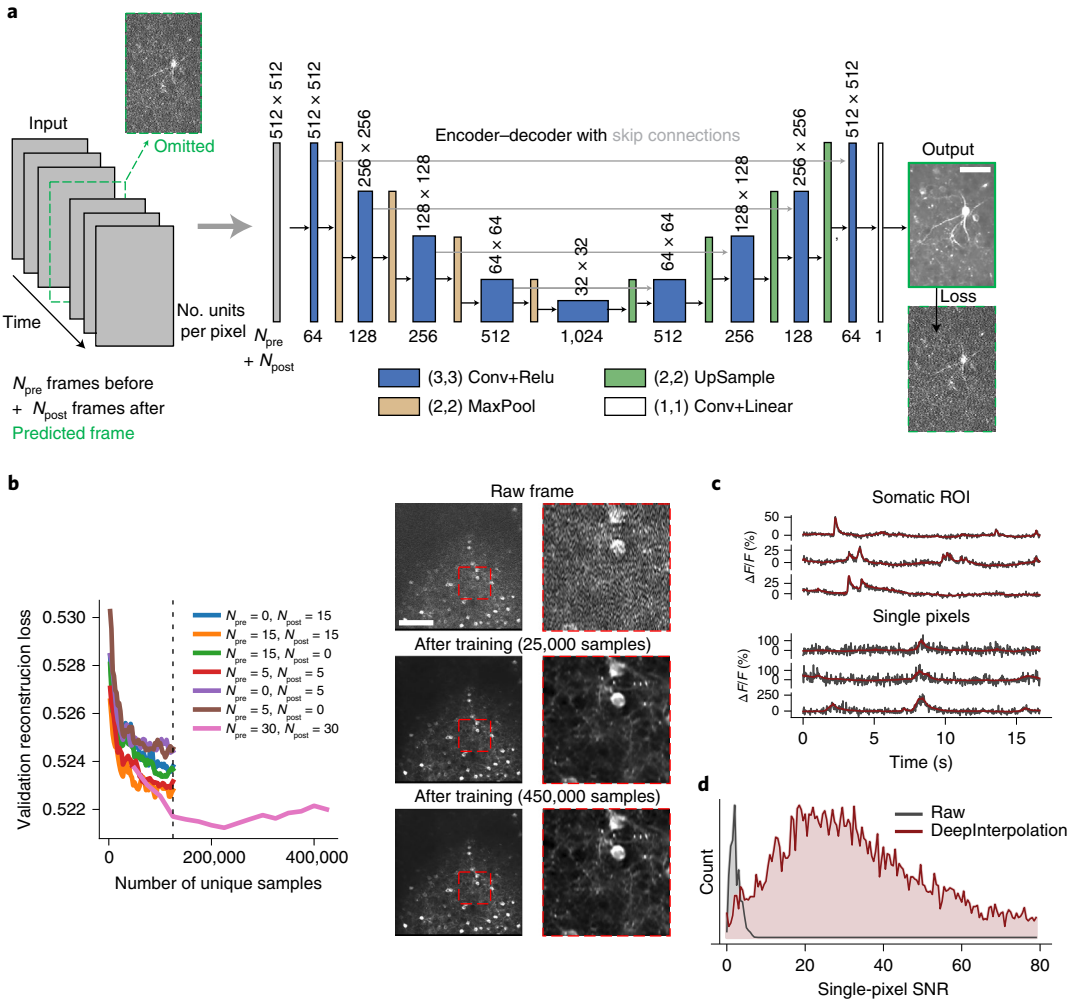


Fig. 1 | Training DeepInterpolation networks for denoising two-photon calcium imaging. a, Schematic of an encoder–decoder deep network with skip connections for denoising two-photon imaging data. The encoder–decoder network used N_{pre} and N_{post} frames acquired at 30 Hz before and after the target frame (labeled in green) to predict the central target frame (which was omitted from the input). The scale bar is 50 µm. Conv refers to 2D convolutional layers, Relu refers to the rectified linear activation function. Linear refers to a linear activation function. UpSample and MaxPool refers to 2D upsampling and 2D downsampling layers. **b**, Left, validation loss shown as a function of unique samples for different combinations of N_{pre} and N_{post} values during training for a single network for the Ai93 reporter line (Methods). The *y* axis is the mean absolute difference between a predicted frame and a noisy sample across 2,500 samples. Individual data samples were *z* scored using a single estimate of mean and s.d. per movie. Validation loss was therefore measured in normalized fluorescence units. The dotted vertical line indicates early stopping (due to extreme computational demands during training) during evaluation of the parameter sets. Right, denoising performance of this model compared to a single raw frame (top) from a single representative experiment. The scale bar is 100 µm. Each red inset is 100 µm in length. **c**, Six representative example traces extracted from a held-out denoised movie before (black) and after (red) denoising with DeepInterpolation. The top three traces were extracted from a somatic ROI, while the bottom three traces were extracted from a single pixel. **d**, Distribution of the SNR (mean over s.d., Methods) for 10,000 pixels (randomly selected across $N = 19$ denoised held-out test movies) before and after application of DeepInterpolation.

(s.e.m.)[7]) suggest that 'frames' (sets of simultaneously acquired two-dimensional (2D) images) up to 1 s away can carry meaningful information.

We trained a neuronal network to reconstruct a center frame from the data present in N_{pre} prior and N_{post} subsequent frames. We omitted the center frame from the input to eliminate any information about the independent pixel-wise shot noise (Fig. 1a). We selected the final values of the meta-parameters ($N_{pre} = 30$ and $N_{post} = 30$) by comparing the validation loss during training (Fig. 1b and Supplementary Note).

Allen Brain Observatory datasets[10] offered a large corpus of data for training our denoising network: for each GCaMP6 reporter line (Ai93 and Ai148), we had access to more than 100 million frames curated with standardized quality-control metrics[10]. Within the limits of our computational infrastructure, we showed that using a larger number of both prior and subsequent frames as input yielded smaller validation reconstruction losses (Fig. 1b). Training required 225,000 samples pulled randomly from 1,144 separate 1-h-long experiments before improvement stopped (Fig. 1b). As the loss was dominated by independent noise present in the target image, the reconstruction loss could not converge to zero. This was supported by simulating ideal reconstruction losses with known ground truth (Supplementary Fig. 2). In fact, even small improvements in the loss were associated with visible improvements in the reconstruction quality of the signal (Fig. 1b). At a typical frame rate of 30 Hz, a single hour of imaging yields 108,000 samples. Therefore, an average laboratory should have sufficient data to train their own DeepInterpolation network.

Applying this trained network to denoise held-out datasets yielded remarkable improvements in image quality (Supplementary Videos 1 and 2). The same trained network generalized well to additional examples (Supplementary Video 2). Shot noise was visibly eliminated from the reconstruction (Fig. 1c), while calcium dynamics were preserved, yielding a 15-fold increase in the SNR (Fig. 1d, mean raw pixel SNR = 2.4 ± 0.01 (s.e.m.), mean pixel SNR after DeepInterpolation = 37.2 ± 0.2 (s.e.m.), N = 9,966 pixels). While movies used during training were corrected for motion, close inspection of the reconstructions showed that DeepInterpolation automatically removed small remaining motion artifacts (Supplementary Videos 1 and 2).

We compared our denoising approach to penalized matrix decomposition (PMD)[11]. While PMD properly reconstructed somatic activities, unlike DeepInterpolation, it rejected most of the variance present in background pixels (Supplementary Video 3). A key underlying assumption of PMD is the absence of motion artifacts. Contrary to what we saw after denoising with DeepInterpolation, we observed artifacts in movies denoised with PMD due to small residual motion artifacts (arrow in Supplementary Video 3). To compare these two approaches while considering this limitation of PMD, we created simulated datasets devoid of motion artifacts, for which we had knowledge of the underlying ground truth signals, using the in silico neural anatomy and optical microscopy (NAOMi) approach[12]. NAOMi combines a detailed anatomical model of a volume of tissue with models of light propagation and laser scanning to generate realistic calcium imaging datasets (Extended Data Fig. 1 and Supplementary Video 4). Because DeepInterpolation used information from nearby frames, it provided smoother calcium traces with reconstruction losses 1.4–2.3 times smaller than those from PMD (Extended Data Fig. 1). We confirmed that DeepInterpolation was better at rejecting high-frequency noise (Fourier transforms in Extended Data Fig. 1). We also compared DeepInterpolation with the pixel-wise approach of Noise2Void[6] as well as with a temporal Gaussian kernel (Supplementary Video 5). As with PMD, the reconstruction error was four times higher with Noise2Void than that with DeepInterpolation (Supplementary Fig. 3d). While the Gaussian kernel was effective at rejecting shot noise, it heavily impacted the calcium event shape due to spatiotemporal smoothing effects (Extended Data Fig. 2). These results support the benefits of training more complex spatiotemporal models.

It was crucial to ensure that DeepInterpolation faithfully reconstructs calcium dynamics without adding any spurious transients, as deep learning models are capable of 'overinterpretation' or enhancing features that do not actually exist[13]. To address this concern, we used our simulated data to compare individual reconstructed calcium transients to those in the ground truth dataset (Extended Data Fig. 2a). First, unlike after Gaussian smoothing, spike-triggered averages showed remarkably accurate reconstruction of individual calcium transients, maintaining the peak amplitude and the area under the events (Extended Data Fig. 2b,c). Second, we leveraged an existing dataset in which both calcium activity and action potentials were recorded simultaneously from the same neurons (Extended Data Fig. 3a,b)[14]. The relationship between action potentials and calcium transients can be complex; therefore, we selected a subset of recordings performed in a Camk2a-tTA;tetO-GCaMP6s transgenic mouse in which this relationship was strong (lower false positive rate in raw data). We found that DeepInterpolation did not alter the amplitude of calcium events associated with one or more action potentials (Extended Data Fig. 3c). Furthermore, DeepInterpolation decreased the false positive rate of event detection without affecting the hit rate (Extended Data Fig. 3d). Smaller bin sizes increased this effect (down to 7 ms for the original sampling rate of 150 Hz), demonstrating that DeepInterpolation can improve the temporal precision of spike inference.

DeepInterpolation transforms segmentation of active compartments. We next evaluated the impact of DeepInterpolation on the segmentation of neuronal compartments in movies of calcium activity. Existing approaches to segmentation rely on the analysis of correlation between pixels to create regions of interest (ROIs)[15,16]. After denoising, single-pixel pairwise correlations greatly increased from near zero (average Pearson correlation = $9.0 \times 10^{-4} \pm 2.3 \times 10^{-6}$ (s.e.m.), $n = 4 \times 10^8$ pairs of pixels across four experiments) to a significantly positive value (average Pearson correlation = 0.10 ± 1.210^{-5} (s.e.m.), $n = 4 \times 10^8$ pairs of pixels across four experiments; Kolmogorov–Smirnov test comparing raw data with DeepInterpolation results, $P = 9 \times 10^{-71}$, n = 1,000 pixels randomly subselected, Supplementary Fig. 3). We expected that this increase in average correlation would improve the quality and the number of segmented regions. To this end, we tested its impact on segmentation algorithms from Suite2p[16] and CaImAn[17,18], two widely used calcium analysis packages.

The improvement in correlation structure provided by DeepInterpolation benefited both algorithms. We first compared cell filters (images representing the weighted distribution of the compartments) with and without DeepInterpolation (Supplementary Figs. 4 and 5). With Suite2p, neurons with higher SNR had identical filters, while dimmer neuronal compartments showed better-defined filters with DeepInterpolation, sometimes recovering missing portions of somas (Supplementary Fig. 4). DeepInterpolation had a similar impact on CaImAn (Supplementary Fig. 5). We leveraged synthetic datasets generated with NAOMi[12] to highlight the role of key segmentation parameters in weighting true positive and false positive rates of individual filters after application of DeepInterpolation (Extended Data Fig. 4). From real data, Suite2p extracted additional cell filters (Fig. 2a), sometimes encompassing dendrites or axons connected to a local soma or even boutons attached to a horizontal dendrite (Fig. 2b). In some movies, smaller sections of axonal or dendritic compartments were detected, each with well-defined calcium transients (Fig. 2c). In future experiments, it might be helpful to acquire high-quality structural data to better define these small compartments.

Although some compartments were not detected in the original raw movie, we applied weighted masks to both denoised and raw movies to quantify the improvement in the SNR of the associated temporal trace. ROIs associated with somatic compartments typically yielded a mean SNR of 21.0 ± 0.5 (s.e.m., n = 240 in three experiments) in the original movie (Fig. 2d); after denoising, most of these ROIs had an SNR above 40 (mean, 73.8 ± 0.5 (s.e.m.)). Similarly, the mean SNR across all compartments, including the smallest apical dendrites, went from a mean of 13.2 ± 0.1 (s.e.m., n = 3,385 in three experiments) to 73.8 ± 0.5 (s.e.m.). To detect an increase in the calcium signal of about 10%, an SNR above 20 is necessary for those events to be twice as large as the noise. This suggests that removing shot noise may increase our capability to detect the presence of single spikes. Detecting single spikes remains difficult owing to the complexity of the relationship between calcium levels and action potentials[14,19], which is unrelated to the presence of independent noise. Nonetheless, the average number of detected active ROIs with an SNR above 20 went from 178 ± 135 (s.e.m., in three experiments) to 1,122 ± 371 in a 400 × 400-μm² field of view (Fig. 2d); that is, six times more neuronal compartments became available for analysis from a single movie following application of DeepInterpolation.

DeepInterpolation improves the analysis of correlated activity. We next analyzed the response of all segmented compartments to ten repeats of a natural movie presentation. We found examples of both large and small responses to natural movies in somatic and non-somatic compartments (Extended Data Fig. 5). The trial-by-trial response reliability increased substantially from

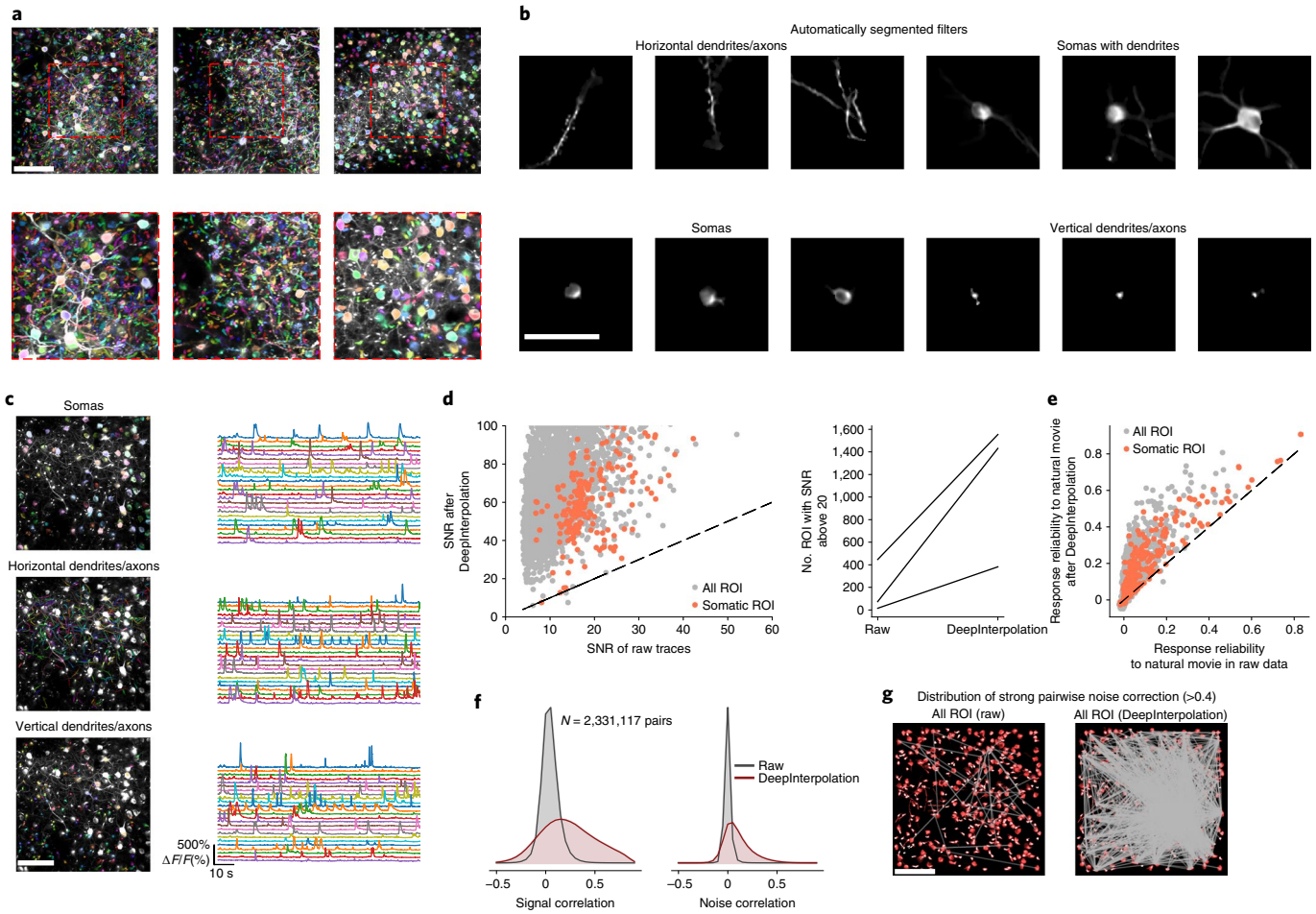


Fig. 2 | Applying DeepInterpolation to calcium imaging reveals additional segmented ROIs and rich network physiology across small and large neuronal compartments. a, Three examples showing overlaid colored segmentation masks on top of a maximum-projection image. Red dashed boxes show zoomed-in views. The scale bar is 100 µm. **b**, Example individual weighted segmented filters showcasing dendrites, isolated somas, somas with attached dendrites and axons and small sections of dendrites or axons. The scale bar is 50 µm **c**, Manually sorted ROIs from one experiment showcasing calcium traces extracted from each individual type of neuronal compartment (from top to bottom, cell body, horizontal processes, processes perpendicular to the imaging plane). **d**, Left, quantification of the SNR for all detected ROIs (gray dots) and somatic ROIs (red dots) with and without application of DeepInterpolation. The dashed line represents the identity line. Right, comparison of the number of ROIs with an SNR above 20 with and without application of DeepInterpolation. Each line is a single experiment ($n = 3$ experiments). **e**, Quantification of the response reliability of individual ROIs across ten trials of a natural movie visual stimulus. The dashed line represents the identity line. **f**, Left, signal correlation (average correlation coefficient between the average temporal response of a pair of neurons) for all pairs of ROIs in **d** for both raw and denoised traces. Right, noise correlation (average correlation coefficient at all time points of the mean-subtracted temporal response of a pair of neurons) for all pairs of ROIs in **d**. **g**, For an example experiment, pairs of ROIs with high noise correlation (>0.4) are connected with a straight line for both original two-photon data and after application of DeepInterpolation. The scale bar is 100 µm.

raw traces to traces after application of DeepInterpolation (Fig. 2e) both for somatic ROIs (raw, mean, 0.10 ± 0.01; DeepInterpolation, mean, 0.20 ± 0.01; s.e.m., $n = 240$ ROI in $n = 3$ experiments) and across all ROIs (raw, mean, 0.029 ± 0.001; DeepInterpolation, mean, 0.113 ± 0.002; s.e.m., $n = 3{,}385$ ROIs in $n = 3$ experiments).

Neuronal representations are inherently noisy. This property of neuronal networks has fueled studies on the relationship between the average signal representation in individual neurons and the trial-by-trial fluctuation of their responses[20–22]. Experimental noise impacts this measure of neuronal relationships as it modulates trial-by-trial responses. For each pair of ROIs in a single experiment, we extracted their shared signal and noise correlation. As expected, we found that both signal and noise correlation increased after denoising (signal, from 0.06725 ± 0.00006 (s.e.m.) to 0.25 ± 0.0001 (s.e.m.); noise, from 0.02240 ± 0.00002 (s.e.m.) to 0.12351 ± 0.00008 (s.e.m.; both for 2,329,554 pairs of ROIs from three experiments; Fig. 2f). This result was not solely due to improved segmentation, as it was preserved using ROIs only detected in raw noisy data (Supplementary Fig. 6). To further illustrate how these improved measures of neuronal fluctuations impacted the analysis of functional interactions, we compared the spatial distribution of pairwise correlations between neurons during natural movie presentation. DeepInterpolation largely increased the number of strong pairwise interactions for both somatic and non-somatic pairs (Fig. 2g, all ROIs, 43–1,721 pairs), including pairs of ROIs much farther apart than our DeepInterpolation model's receptive field of 60 µm.

Fine tuning generalizes pre-trained models to previously unseen datasets. How well will this approach scale beyond the highly standardized datasets that we have analyzed thus far? We originally aggregated training data across hundreds of experiments to generate a few DeepInterpolation networks that could generalize across

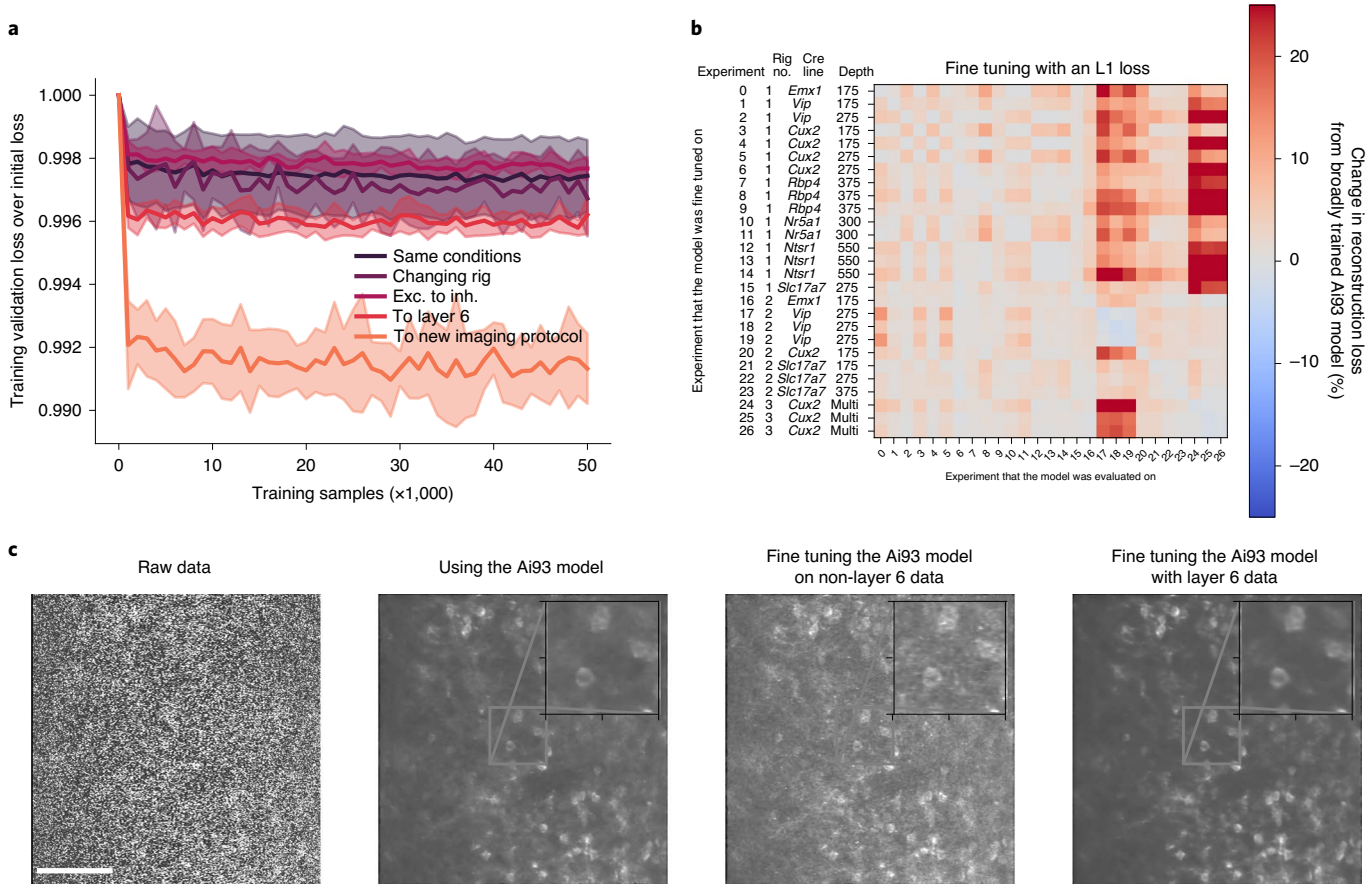


Fig. 3 | Fine tuning pre-trained DeepInterpolation models recovers reconstruction performance with minimal training data. a, Training reconstruction loss in validation data throughout fine tuning of a pre-trained model (Ai93 DeepInterpolation model) on previously unseen two-photon calcium movies. For training, experimental conditions differed as follows: switching to a different experimental rig with the same recording conditions (changing rig), using data from inhibitory neurons (excitatory (exc.) to inhibitory (inh.)) recorded with a different reporter line (Ai148), using data from layer 6 (to layer 6) recorded with an *Ntsr1*-Cre_GN220;Ai148 mouse line, using data from the same mouse line but imaged at 10 Hz (instead of 30 Hz) and performing volume scanning with a piezo objective scanner (to new imaging protocol). Dashed lines represent the s.d. across rounds of training with similar conditions. **b**, Cross-model evaluation performance when varying recording conditions. The broadly trained Ai93 DeepInterpolation model was fine tuned separately on 26 recording experiments with varying imaging conditions. We then evaluated each fine tuned model on all 26 experiments separately to plot cross-condition reconstruction performance. Performance was normalized to the broadly trained Ai93 model. **c**, Comparison of an example raw frame recorded with an *Ntsr1*-Cre_GN220;Ai148 mouse line in layer 6 (left) with its corresponding frame after application of DeepInterpolation with various models. First, we used a broadly pre-trained Ai93 model and then a fine tuned model on non-layer 6 data and finally a model trained on the same exact layer 6 experiment. The scale bar is 100 µm.

different recording conditions. These broadly trained models should be a good starting point for fine tuning on other datasets, provided that individual frames of the data can fit within our 512-by-512 window, with unoccupied pixels filled with zero values.

To test this, we selected 26 datasets collected with both different and shared experimental conditions: one subset included mice with labeled inhibitory neurons instead of excitatory neurons; another subset was collected using a different model of microscope; and a final set used an imaging frame rate of 10 Hz (instead of 30 Hz). For each experiment, we fine tuned our pre-trained model using 50,000 random frames. In all cases, training converged after 20,000 frames (Fig. 3a). When evaluating the performance of all 26 models on all pairwise movie combinations, we found that fine tuned models had improved reconstruction performance at the expense of generalization across experimental conditions (Fig. 3b). We applied DeepInterpolation to recordings from neurons in cortical layer 6, which have higher levels of background fluorescence than recordings from more superficial layers (Fig. 3c). Improvements were especially visible in the reconstruction of the background (Fig. 3d and Supplementary Video 6). When fine tuning for low-SNR experiments (below five photons per pixel per dwell time), we found that an L1 loss could cause abnormally damped fluorescence values (Extended Data Fig. 6). While L1 loss forces training to converge toward the median of all potential outcomes, providing resilience to artifactual frames, such as motion artifacts, it discards the most meaningful output frames at low photon count. In this case, fine tuning with an L2 loss, which converges toward the mean of all potential outcomes, recovered denoising performances with low photon-count experiments (Extended Data Fig. 6 and Supplementary Video 7).

Application to electrophysiology. Electrophysiological recordings from high-density silicon probes have characteristics similar to those of two-photon imaging movies: information is shared across nearby pixels (electrodes) as well as across time. Thus, a similar architecture should perform well for denoising electrophysiology data, in particular, for removing strictly independent thermal and shot noise from recordings.

Because a single action potential lasts ~1 ms, we constructed our interpolating network to predict the voltage profile at any given

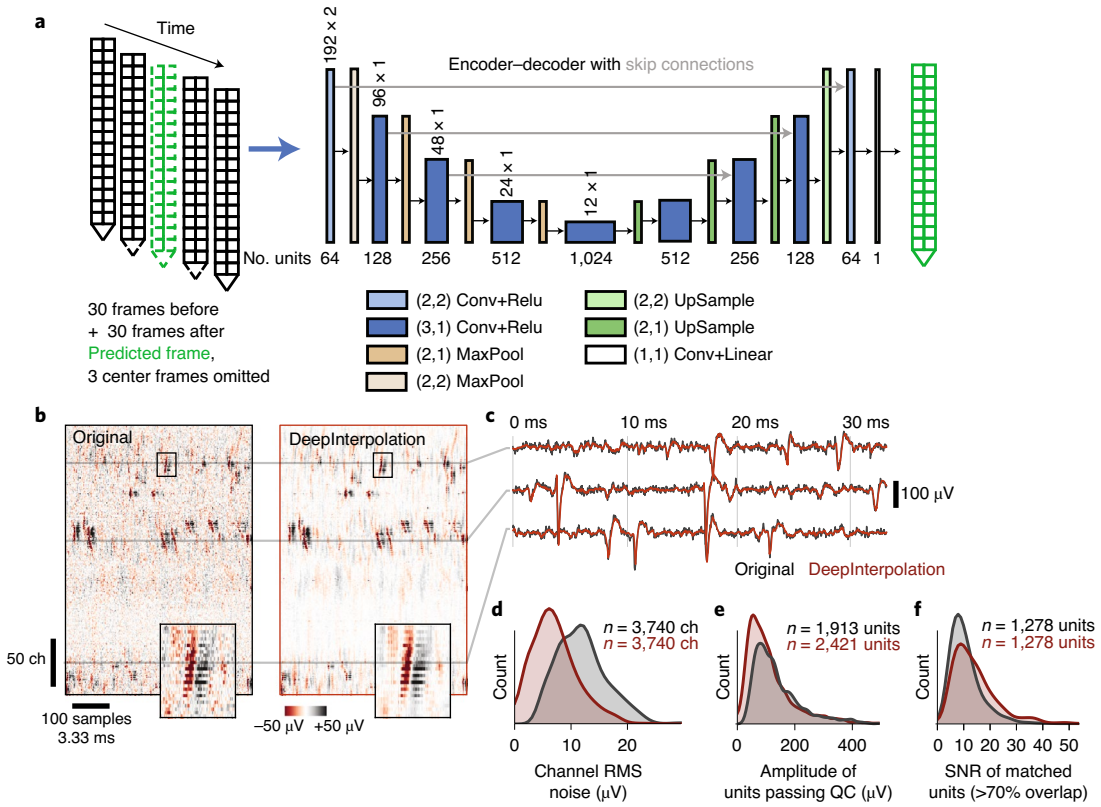


Fig. 4 | Applying DeepInterpolation to electrophysiological recordings decreases background noise and yields additional detected neuronal units. a, Structure of the DeepInterpolation encoder–decoder network for electrophysiological data recorded from a Neuropixels silicon probe with two columns of 192 electrodes each, spaced 20 µm apart. Data were acquired at 30 kHz. **b**, Side-by-side comparison of original (left) and denoised (right) data, plotted as a 2D heatmap with time on the horizontal axis and channels (ch) along the vertical axis. Insets show a close-up of a single action potential across 22 contiguous channels (the box is 3.6 ms wide). **c**, Denoised time series (red) overlaid on the original time series (gray) for three channels from **b**. **d**, Histogram of RMS noise for all channels from ten experiments, before and after applying DeepInterpolation. **e**, Histogram of waveform amplitudes for all high-quality units from ten experiments. QC, quality control. **f**, Histogram of waveform SNRs for all units that were matched before and after denoising.

moment from 30 preceding and 30 following data points acquired at 30 kHz. Unlike for two-photon imaging, we found that additionally omitting one sample before and after the center sample allowed better rejection of the background noise while maximizing signal-reconstruction quality. The architecture of the network was similar to the one used for two-photon imaging, except that we reshaped the input layer to match the approximate layout of the Neuropixels recording sites (Fig. 4a). Because electrophysiological recordings have a sampling rate 1,000-fold higher than that of two-photon imaging, we trained the network on only three experiments in total[23,24].

After inference, a qualitative inspection of recordings showcased excellent noise rejection (Fig. 4b). We compared spike profiles before and after denoising and found more clearly visible spike waveforms in denoised data, especially for units of lower amplitude (Fig. 4b, inset, and Fig. 4c). The residual of our reconstruction showed weak spike structure, but spectral analysis revealed a mostly flat frequency distribution (Extended Data Fig. 7a). The overall shape (amplitude and decay time) of individual action potentials was preserved, while the median channel root mean square (RMS) noise was decreased 1.7-fold (6.96 µV after denoising, Fig. 4d). After denoising and spike sorting, we found 25.5% ± 14.5% (s.d.) more high-quality units per probe (inter-spike interval violation score < 0.5, amplitude cutoff < 0.1, presence ratio > 0.95). The number of detected units was higher after applying DeepInterpolation regardless of the chosen quality metric thresholds (Extended Data Fig. 7b). Most additional units had low-amplitude waveforms, with the number of units with amplitudes above 75 µV remaining roughly the same (1,532 units before versus 1,511 units after), while the number with amplitudes below 75 µV increased (381 units before versus 910 units after) (Fig. 4e). For units that were matched before and after denoising (at least 70% overlapping spikes), the average SNR increased from 11.2 to 14.5 (Fig. 4f).

By searching for spatiotemporally overlapping spikes, we determined that 6.2% ± 2.9% (s.d.) of units in the original recording were no longer detected after applying DeepInterpolation (<50% matching spikes). However, this was counterbalanced by the addition of 20.2% ± 4.8% (s.d.) units that had fewer than 50% matching spikes in the original data (Extended Data Fig. 7c). This is less than the 25.5% increase cited above, because it accounts for a minority of units that were detected but merged in the original data. To validate that these additional units were likely to correspond to actual neuronal compartments, we analyzed their responses to natural movies using the same movie as that in the two-photon imaging study. We found approximately 9% more reliably visually modulated units in the visual cortex after denoising (Extended Data Fig. 7d) and found examples of stimulus-modulated units that were previously undetected (Extended Data Fig. 7e). These results showcase the ability of DeepInterpolation to reveal additional biologically relevant information about the neuronal circuits under investigation.

We compared the impact of DeepInterpolation on our electrophysiology data to that of two much simpler denoising methods, smoothing with a Gaussian or boxcar window (Extended Data Fig. 8a). Both simpler methods were able to reduce RMS noise

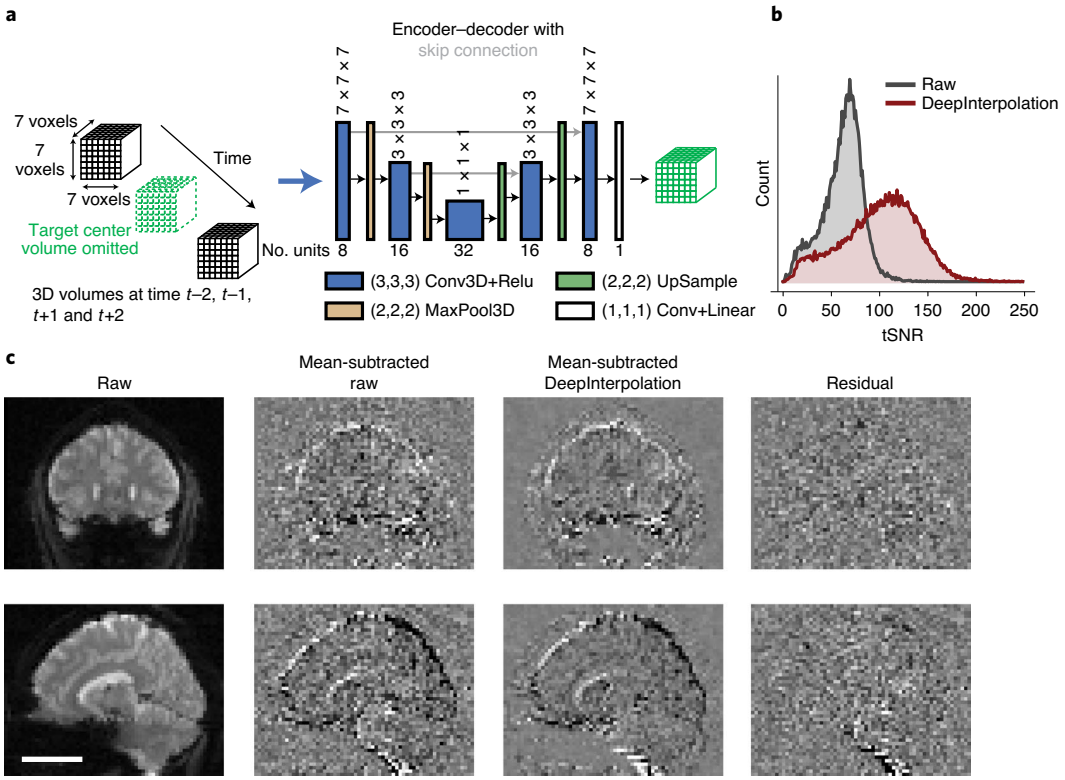


Fig. 5 | Applying DeepInterpolation to fMRI removes thermal noise. a, Structure of the DeepInterpolation encoder–decoder network for fMRI data. Instead of predicting a whole brain volume at once, the network reconstructs a local 7×7×7 cube of voxels. **b**, The temporal SNR for 10,000 voxels randomly distributed in the brain volume in raw data and after applying DeepInterpolation. **c**, Exemplar reconstruction of a single fMRI volume. The first row is a coronal section, while the second row is a sagittal section through a human brain. In the second column, the temporal mean of a 3D scan was removed to better illustrate the presence of thermal noise in the raw data. The local denoising network was processed throughout the whole 3D scan for denoising. The impact of DeepInterpolation on thermal noise is illustrated in the third column. The fourth column shows the residual of the denoising process. The scale bar is 5 cm.

levels to a range consistent with that achieved by DeepInterpolation (Extended Data Fig. 8b). However, this came at the expense of spike amplitude attenuation (Extended Data Fig. 8c,d).

To determine the conditions under which DeepInterpolation can preserve spike waveform shape, we applied it to a simulated ground truth dataset in which spikes were superimposed at random locations in the original data (Extended Data Fig. 9). For most waveforms and amplitudes, waveform shape was preserved after applying DeepInterpolation. However, we did observe some attenuation at low amplitudes and for narrow waveforms (full width at half-maximum < 0.15 ms). Reduced similarity is expected at low amplitudes, at which noise fluctuations can cause distortions in waveform shape. Future iterations of the DeepInterpolation approach could mitigate this by using a similarity metric for narrow spike waveforms in simulated ground truth datasets to iterate the network architecture.

Application to fMRI. We evaluated whether DeepInterpolation could aid analysis of volumetric datasets such as those from fMRI. fMRI is very noisy, as the BOLD response is typically only a 1–2% change of the total signal amplitude[25]. Thermal noise is present in the electrical circuit receiving the magnetic resonance signal. A typical fMRI-processing pipeline involves averaging nearby pixels and successive trials to increase the SNR[26,27]. Thus, our approach could replace smoothing kernels with more optimal local interpolation functions without sacrificing spatial or temporal resolution.

Given the lower spatiotemporal sampling rate of fMRI, we trained a more local interpolation function with only two pooling layers. To reconstruct the value of a brain sub-volume, we fed a neural network with a consecutive temporal series of a local volume of 7×7×7 voxels. We omitted the entire target volume from the input (Fig. 5a). We trained a single denoising network using 140 fMRI datasets acquired across five individuals, comprising 1.2 billion samples in total.

We extracted temporal traces from individual voxels and found that the SNR increased from 61.6 ± 20.6 (s.d.) to 100.40 ± 38.7 (s.d.) (n = 100,001 voxels) (Fig. 5b). Applying the trained network to held-out test data resulted in excellent denoising performance (Fig. 5c). We noticed that the background noise around the individual's head was excluded and only present in the residual image (Fig. 5c and Supplementary Video 8). Surrounding soft tissues became visible after denoising (Fig. 5c, raw data, background voxel standard deviation = 7.59 ± 0.01 (s.e.m.); brain voxel std = 15.95 ± 0.08 (s.e.m.), n = 10,000 voxels; DeepInterpolation data, background standard deviation = 2.24 ± 0.01 (s.e.m.), n = 10,000; brain voxel standard deviation = 9.72 ± 0.05 (s.e.m.), n = 10,000 voxels; residual data, background standard deviation = 7.91 ± 0.01; brain voxel standard deviation = 15.79 ± 0.08, n = 10,000 voxels). We compared movies extracted with DeepInterpolation to fully preprocessed movies smoothed with a Gaussian denoising kernel (Supplementary Video 9): the original three-dimensional (3D) resolution was fully maintained after DeepInterpolation, while processing data with a Gaussian kernel diffused dynamics across nearby voxels. The residual movie showed no visible structure except for occasional blood vessels, corrected motion artifacts and head-mounting hardware (Fig. 5c and Supplementary Video 10).

Discussion

We have presented DeepInterpolation, which can reconstruct nearly noiseless versions of datasets without the need for ground truth training data. Applying DeepInterpolation to calcium imaging, electrophysiological recordings and fMRI demonstrates its widespread benefits and ability to uncover previously hidden neuronal dynamics. While our approach is closely related to Noise2Self[5] and Noise2Void[6], we developed our framework independently. As a result, several key differences are notable. First, instead of working with single fluorescence frames with pixel-wise omissions, we adapted our approach to the types of complex multidimensional biological datasets at the heart of systems neuroscience. Second, we trained our models on large databases, demonstrating the impact of richer denoising models on existing neuroscience data and on workflows such as cell segmentations or spike sorting. Third, we contributed practical solutions to relevant problems in this domain: we provide a network topology that works on real neuroscience datasets, a training regimen that requires data volumes that are available to most laboratories and a fine tuning process to adapt pre-trained models to new datasets.

We anticipate that DeepInterpolation will enhance post-processing steps that can leverage higher SNR levels and more pronounced correlation structure. For instance, background correction in two-photon imaging data relies on measurements of the contamination from neighboring pixels. Typically, measurements within background pixels are less reliable than those in somatic pixels, given the lower expression level in the background, necessitating averaging of distant pixels. With fewer, more local neighboring pixels needed to make this measurement, we expect background-correction reliability to increase.

We anticipate that DeepInterpolation will become instrumental for the advancement of large-scale voltage imaging of neuronal activity[28,29] and will facilitate practical application of high-speed fMRI, which currently operates in a regime in which thermal noise dominates the BOLD signal. Our approach should permit scientists in a variety of fields to reanalyze their existing datasets but with independent noise reduced or removed.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-021-01285-2.

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Methods

Description of experimental data used in this study. All experimental datasets used in this study were previously collected and published separately. Associated datasets were all made available online. Animal and human work approval was described in each associated publication.

We trained four denoising neuronal networks in this study: one for two-photon imaging experiments using the Ai93(TITL-GCaMP6f) reporter line, one for the Ai148(TIT2L-GCaMP6f-ICL-tTA2) reporter line[30,31], one for Neuropixels recordings using 'phase 3a' probes[32] and one for fMRI imaging using datasets from a study published on https://openneuro.org/[33,34]. For Neuropixels, training data consisted of 1-h segments extracted from ten different ~2.5-h-long recordings (see below for information on the specific sessions used). For fMRI, the dataset contained fMRI data from five individuals with three types of scanning sessions: 'ses-perceptionTraining', 'ses-perceptionTest' and 'ses-imageryTest'. We trained our denoiser on 'ses-perceptionTraining' sessions and measured the denoising performance on 'ses-perceptionTest' sessions.

All raw datasets used for training are available at the time of publication through AWS S3 buckets (https://www.amazon.com/, https://openneuro.org/datasets/ds001246/versions/1.2.1 and https://registry.opendata.aws/allen-brain-observatory/).

Preprocessing steps. Two-photon imaging movies were corrected for motion similarly to the method in our previous publication[10]. The motion-correction algorithm relied on phase correlation and only corrected for rigid translational errors. We used these motion-corrected movies for denoising. All motion-corrected movies were used directly as they were during training; we did not exclude any frames or recordings due to motion instabilities. The SNR was not used as a criterion for inclusion and was highly variable in the training set, as we usually set laser power to the maximum level safely tolerated by the tissue[10]. Potential motion-correction errors occasionally present in our data were typically below neuronal soma size. We did not reapply motion correction after DeepInterpolation.

For Neuropixels recordings, the median across channels was subtracted to remove common-mode noise. The median was calculated across channels that were sampled simultaneously, leaving out adjacent channels that were likely measuring the same spike waveforms as well as reference channels that contain no signal. For each sample, the median value of channels N:24:384, where $N = (1,2,3,\ldots,24)$, was calculated, and this value was subtracted from the same set of channels. Importantly, this step removes noise that is correlated across channels without affecting the independent noise targeted by DeepInterpolation. We performed denoising on the spike band after median-subtraction and offline filtering steps (150 Hz, high pass) were applied.

For fMRI recordings, we used raw, unprocessed NIfTI-1 volume data provided at https://openneuro.org/datasets/ds001246/versions/1.2.1. We compared denoised fMRI recordings with processed datasets available at https://openneuro.org/ that included a Gaussian kernel. The Gaussian kernel was part of a standard fMRI-processing pipeline using SPM5 (http://www.fil.ion.ucl.ac.uk/spm) and was introduced as part of a motion-correction algorithm. We did not alter the parameters of this Gaussian kernel.

Detection of ROIs in two-photon data. We used two different segmentation algorithms in this study. The first one was described in a previous publication and leverages a succession of morphological filters to extract binary masks surrounding active pixels[10]. These masks are publicly available through the Allen SDK (https://allensdk.readthedocs.io/en/latest/). In all analyses related to Supplementary Fig. 6, we applied these binary masks to both raw and DeepInterpolation movies to extract matching calcium traces.

In all analyses related to Fig. 2, we used the sparse mode in Suite2p[16] to extract all individual filters. We used default parameter values of this mode except for 'threshold_scaling', which was set to 3. With the increased SNR achieved with DeepInterpolation, this single change limited the proportion of false positives in the final set of filters (Extended Data Fig. 4e). Using the Suite2p sorting GUI, we then manually sorted filters. We excluded filters present at the edge of the image and filters created by motion artifacts as well as features that did not cover any neuronal segment present in the maximum-projection image. Somatic and non-somatic ROI were also manually sorted based on the presence of a soma-excluded region in the filter. Filters that included a blurred out-of-plane soma were considered non-somatic. Once the final set of filters were extracted, we reapplied those weighted masks either to the original raw movie or to the movie after applying DeepInterpolation to extract individual traces.

For both Supplementary Figs. 4 and 5, we used default parameters in Suite2p[16] and CaImAn[17] and matched cell filters using a Euclidean distance measure in each filter set.

Detection of neuronal units in electrophysiological data. Kilosort2 was used to identify spike times and assign spikes to individual units[35]. Traditional spike-sorting techniques extract snippets of the original signal and perform a clustering operation after projecting these snippets into a lower-dimensional feature space. By contrast, Kilosort2 attempts to model the complete dataset as a sum of spike 'templates.' The shape and location of each template is iteratively refined until the data can be accurately reconstructed from a set of N templates at M spike times with each individual template scaled by an amplitude, a.

Kilosort2 was applied to the original and denoised datasets using identical parameters (all default parameters, except for ops.Th, which was lowered from (10 4) to (7 3) to increase the probability of detecting low-amplitude units). Because the spike-detection threshold is relative to the overall noise level per channel, the absolute value of the threshold was lower following application of DeepInterpolation.

The Kilosort2 algorithm will occasionally fit a template to the residual left behind after another template has been subtracted from the original data, resulting in double-counted spikes. This can create the appearance of an artificially high number of ISI violations for one unit or artificially high zero-time-lag synchrony between nearby units. To eliminate the possibility that this artificial synchrony will contaminate data analysis, the outputs of Kilosort2 are post-processed to remove spikes with peak times within five samples (0.16 ms) and peak waveforms within five channels (~50 µm).

Kilosort2 generates templates of a fixed length (2 ms) that matches the time course of an extracellularly detected spike waveform. However, there are no constraints on template shape, which means that the algorithm often fits templates to voltage fluctuations with characteristics that could not physically result from the current flow associated with an action potential. The units associated with these templates are considered 'noise' and are automatically filtered out based on three criteria: spread (single channel or >25 channels), shape (no peak and trough, based on wavelet decomposition) or multiple spatial peaks (waveforms are non-localized along the probe axis). A final manual inspection step was used to remove any additional noise units that were not captured by the automated algorithm.

Quality control for electrophysiological units. Units with action-potential-like waveforms detected by Kilosort2 are not necessarily of high quality. To ensure that units met basic quality standards for further analysis, we filtered them using three different quality metrics as follows, computing with the 'ecephys_spike_sorting' Python package (https://github.com/AllenInstitute/ecephys_spike_sorting). (1) ISI violation score < 0.5. This metric is based on refractory period violations that indicate that a unit contains spikes from multiple neurons. The ISI violation metric represents the relative firing rate of contaminating spikes. It is calculated by counting the number of violations <1.5 ms, dividing by the amount of time for potential violations surrounding each spike and normalizing by the overall spike rate. It is always positive (or 0) but has no upper bound. See ref. [36] for more details. (2) Amplitude cutoff < 0.1. This metric provides an approximation of a unit's false negative rate. First, a histogram of spike amplitudes is created, and the height of the histogram at the minimum amplitude is extracted. The percentage of spikes above the equivalent amplitude on the opposite side of the histogram peak is then calculated. If the minimum amplitude is equivalent to the histogram peak, the amplitude cutoff is set to 0.5 (indicating a high likelihood that >50% of spikes are missing). This metric assumes a symmetrical distribution of amplitudes and no drift; therefore, it will not necessarily reflect the true false negative rate. (3) Presence ratio > 0.95. The presence ratio is defined as the fraction of blocks within a session that include one or more spikes from a particular unit. Units with a low presence ratio are likely to have drifted out of the recording or could not be tracked by Kilosort2 for the duration of the experiment.

Applying these quality metrics removed 54% of detected units in the original data and 60% of units after denoising. Spike sorting after applying DeepInterpolation revealed more units regardless of the threshold used for each quality-control metric (Extended Data Fig. 7b).

Procedure for finding matched versus new units. Following spike-sorting steps, we searched for overlapping spikes between pairs of units detected before and after denoising. Spikes were considered to overlap if they had a peak occurring within ±5 channels and 0.5 ms of one another. The number of overlapping spikes was used to compute three metrics using the original spike trains as 'ground truth' (ref. [37]): (1) precision (the fraction of denoised spikes that were also found in the original data), (2) recall (the fraction of original spikes that were also found in the denoised data) and (3) accuracy $(N_{\text{match}} \times (N_{\text{denoised}} + N_{\text{original}} - N_{\text{match}})^{-1})$.

Units were considered matched if they had an accuracy exceeding 0.7. Units were considered new after denoising if they had a total precision (summed over all original units) less than 0.5. Units were considered undetected after denoising if they had a total recall (summed over all denoised units) less than 0.5.

Applying smoothing kernels to electrophysiology data. To compare DeepInterpolation to simpler denoising methods, we selected a 10-s chunk of data (300,000 samples × 384 channels) and convolved it with a Gaussian kernel ($\sigma = 1$) or a 4 × 4 boxcar window using methods from the 'scipy.ndimage' library. We then computed RMS noise levels for individual channels after masking out spike waveforms (absolute voltage levels > 50 µV).

Simulated Neuropixels ground truth data. We created simulated ground truth data by superimposing known spike waveforms on a real Neuropixels dataset to quantify any distortions that were present after applying DeepInterpolation. We

first hand-selected 20 mean waveforms with diverse shapes from a separate dataset. These were then scaled between 20 and 300 µV and added to the hour-long dataset at random times and random locations ($N = 100$ spikes per waveform × amplitude combination).

Training of denoising networks. We used three different sets of environments for training deep neural networks. Two-photon denoising networks were trained using TensorFlow 1.9.0, Keras 2.2.4 and CUDA 9.0. Electrophysiology denoising networks were trained using TensorFlow 2.0 and CUDA 10.0 through the built-in Keras libraries. fMRI-denoising networks were trained using TensorFlow 2.2 and CUDA 10.1.

We used NVIDIA Titan X, GeForce GTX 1080 and Tesla V100-SXD2 GPUs available on the Allen Institute internal computing clusters. The fMRI-denoising network was trained on Amazon AWS using p2.8xlarge and p3.8xlarge instance types, depending on availability. For all three data types, training on a single GPU took 2–3 d of continuous time.

We used an L1 loss during training for both two-photon imaging and fMRI datasets. Electrophysiological datasets were trained with an L2 loss. All training was performed with the RMSprop gradient descent algorithm implemented in Keras. The two-photon denoiser was trained with a batch size of 5 to fit on available GPU memory. The learning rate was set to 5×10^{-4}. The Neuropixels denoiser was trained with a batch size of 100, and the learning rate was set to 10^{-4}. The fMRI denoiser was trained with a batch size of 10,000. The larger batch size was allowed by the smaller input–output size. The fMRI denoiser was trained with a learning scheduler and initialized at 10^{-4}, decreasing by half every 45 million samples.

To facilitate training, all samples were mean centered and normalized by a single shared value for each experiment during training. For two-photon movies, the mean and s.d. were precalculated using the first 100 frames of the movie. For Neuropixels recordings, the mean and s.d. were precalculated using 200,000 samples.

For fMRI recordings, we extracted the centered volume of the movie that was one-quarter of the total movie size and averaged all voxels. To allow our interpolation network to be robust to edge conditions, input volumes on the edge of the volume were fed with zeros for missing values. For inference, we convolved the denoising network through all voxels of the volume across both space and time, using only the center pixel of the output reconstructed volume to avoid any potential volume-boundary artifacts.

A detailed step-by-step pseudocode description of this process is available in the Supplementary Note.

Fine tuning of denoising networks. For all datasets in Fig. 3, we started training with the Ai93 excitatory line DeepInterpolation network. In total, 50,000 randomly selected frames were pulled from each individual movie and used only once during training. The validation loss was measured on each movie before training and at the end of training. All data in Fig. 3 were trained using an L1 loss.

Simulation of using an L1 loss versus an L2 loss against different light levels. A single averaged two-photon frame was used as a ground truth to generate 100 noisy frames with Poisson noise added on top. The amplitude of this Poisson noise was determined by the simulated photon count. We randomly inserted black frames to emulate poorly motion-corrected frames before taking either the median or mean of all 100 frames. This median or mean frame was compared to the ground truth frame.

Inference of final datasets with trained networks. Once a DeepInterpolation network was learned, inference was performed by streaming entire experiments through the same fixed network. To match the conditions in training, each dataset was mean centered and normalized following the same procedure used during training. Output denoised data were brought back to their original scale after going through the DeepInterpolation model.

Applying the trained DeepInterpolation network comes with some computational costs. As inference is an 'embarrassingly parallel' problem, we used our internal cluster with more CPU nodes than GPU nodes available for parallelization. On this cluster, one local job with 16 threads available required approximately 0.5–1 s to process one two-photon frame. With 100 jobs simultaneously triggered, we could process an entire 70-min-long recording in 20–60 min, depending on the availability of nodes. Processing individual electrophysiological recordings had a similar computational need.

For quantification of the SNR in two-photon data, the SNR was defined as the ratio of the mean fluorescence value divided by the s.d. along the temporal dimension. In ideal photon shot noise-limited conditions, this SNR is proportional to the square root of N, where N is the photon flux detected per pixel.

In the Neuropixels data, the SNR for individual units was defined as the ratio of the maximum peak-to-peak waveform amplitude to the RMS noise on the peak channel. In the fMRI data, the temporal SNR of individual voxels was defined as the ratio of the mean BOLD signal value divided by the s.d. along the temporal dimension.

Analysis of natural movie responses. *Noise- and signal-correlation analysis in two-photon data.* For each natural movie presentation, we extracted the corresponding ROIs traces. Signal correlation was computed by averaging all ten presentations of the movie and calculating the Pearson pairwise correlation of those averaged responses between pairs of ROIs.

The noise correlation was calculated by calculating the Pearson correlation of individual trial responses between a pair of ROIs and then averaging all ten presentations of the movie.

Response reliability in two-photon data. The response reliability was calculated according to the following formula:

$$\text{reliability(ROI)} = \frac{2}{N(N-1)} \sum_{(i,j)\,i \neq j, i<j} r(\text{response}_{\text{trial}_i}(t), \text{response}_{\text{trial}_j}(t)),$$

where r is the Pearson correlation across time points t between a response trace from an ROI at trial_i and trial_j. N is the total number of trials.

This process yielded a single reliability measure for each ROI.

Significant responses in Neuropixels data. When analyzing spike data, the trial-to-trial correlation of the natural movie response depends heavily on the bin size chosen. To determine responsiveness for Neuropixels data, we instead compared each unit's natural movie PSTH (averaged across 25 trials) to a version in which all bins were temporally shuffled. A Kolmogorov–Smirnov test between the original and shuffled PSTHs was used to determine the probability that the response could have occurred by chance ($P < 0.05$ for significantly responsive units; see Extended Data Fig. 7 for an example).

Synthetic data generation for two-photon imaging. We created realistic synthetic calcium imaging datasets using an approach called in silico NAOMi[12]. Given the computational load of the model, we generated a single dataset made of 15,000 frames simulating a 400×400-µm^2 field of view at a frame rate of 11 Hz. Except for the size of the field of view and frame rate, all parameters were set to default values. We also used three datasets accompanying the NAOMi publication ('sparseLabelingVolume', 'typicalVolume80mW' and 'typicalVolume160mW'). To denoise these synthetic datasets, we fine tuned our Ai93 DeepInterpolation onto the synthetic data using an L2 loss.

Comparison with penalized matrix factorization. PMD[11] was run on an AWS instance ('m5.8xlarge') under a prepackaged JupyterHub environment (https://hub.docker.com/r/paninski/trefide). All movies were pre-centered and normalized using code available in the trefide package ('psd_noise_estimate').

Comparison with Noise2Void and a Gaussian kernel. We used the Fiji[38] implementation of Noise2Void[6,38]. All default parameters (batch size of 64, neighborhood radius of five, patch shape of 64) were used to train a denoising network using a dataset simulated with NAOMi.

To compare DeepInterpolation to a Gaussian kernel, we convolved simulated noisy calcium movies with a Gaussian kernel ($\sigma = 3$) along the temporal dimension.

Comparison of calcium detected events with cell-attached electrophysiological recordings. We used four experiments (cell IDs 102932, 102939, 102941 and 102945) collected in Camk2a-tTA;tetO-GCaMP6s transgenic mice[14]. We first separately fine tuned a DeepInterpolation model to each high-magnification raw movie. With these models, we subsequently denoised each imaging dataset. Cell-attached electrophysiological recordings were not denoised, and we used precomputed spike events provided with each dataset. Raw and denoised movies were processed with Suite2p default parameters, and the recorded cell calcium ROI was manually selected for further processing. Neuropil contamination was corrected using a 0.8 weight, and calcium events were detected using OASIS[39]. Receiver operating characteristic curves were calculated by varying an event-detection threshold on the calcium event traces created by OASIS. Both calcium and detected action-potential traces were binned with varying bin size for comparison. True positive rates were calculated using bins in which one or more detected action potentials were present, while false positive rates were calculated using bins with no detected spikes.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Two-photon imaging and Neuropixels raw data can be downloaded from the following S3 bucket: arn:aws:s3:::allen-brain-observatory. Two-photon imaging files can be accessed according to experiment ID using the following path: visual-coding-2p/ophys_movies/ophys_experiment_>.h5. We used a random subset of 1,144 experiments for training the denoising network for Ai93, and 397 experiments were used for training the denoising network for Ai148. The list of used experiment IDs is available in JSON files (in deepinterpolation/examples/json_data/) on the DeepInterpolation GitHub repository. The majority

of these experiment IDs are available in the S3 bucket. Some experimental data have not been released to S3 at the time of publication. Neuropixels raw data can be accessed by experiment ID and probe ID using the following paths: visual-coding-neuropixels/raw-data/<Experiment ID>/<Probe ID>/spike_band.dat. Median subtraction was applied to the DAT files after processing, but the DAT files do not include an offline high-pass filter. Before applying DeepInterpolation, we extracted 3,600 s of data from each of the recordings listed in Supplementary Table 1. To train the fMRI denoiser, we used datasets that can be downloaded from an S3 bucket: arn:aws:s3:::openneuro.org/ds001246. We trained our denoiser on all 'ses-perceptionTraining' sessions across five individuals (three perception training sessions, ten runs each).

Code availability

Code for DeepInterpolation and all other steps in our algorithm is available online through a GitHub repository[40]. Example training and inference tutorial codes are available at the repository. Code and data to regenerate all figures presented in this study are available online through a GitHub repository[41].

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Author contributions

J.L. initiated, conceptualized and directed the project. J.L. and M.O. conceptualized the DeepInterpolation framework. J.L. wrote all training and inference code with guidance from M.O. J.L. performed all two-photon and fMRI analyses and ran DeepInterpolation on electrophysiological data. J.H.S. and J.L. iterated the electrophysiological denoising process. P.L. wrote code to analyze simultaneous optical and electrophysiological recordings. J.H.S. performed all analysis of electrophysiological data. C.K. guided the characterization of all denoising performances. N.O. ran the two-photon ground truth simulation. J.L., M.O., J.H.S., P.L. and C.K. wrote the paper.

Competing interests

The Allen Institute has applied for a patent related to the content of this study.

Additional information

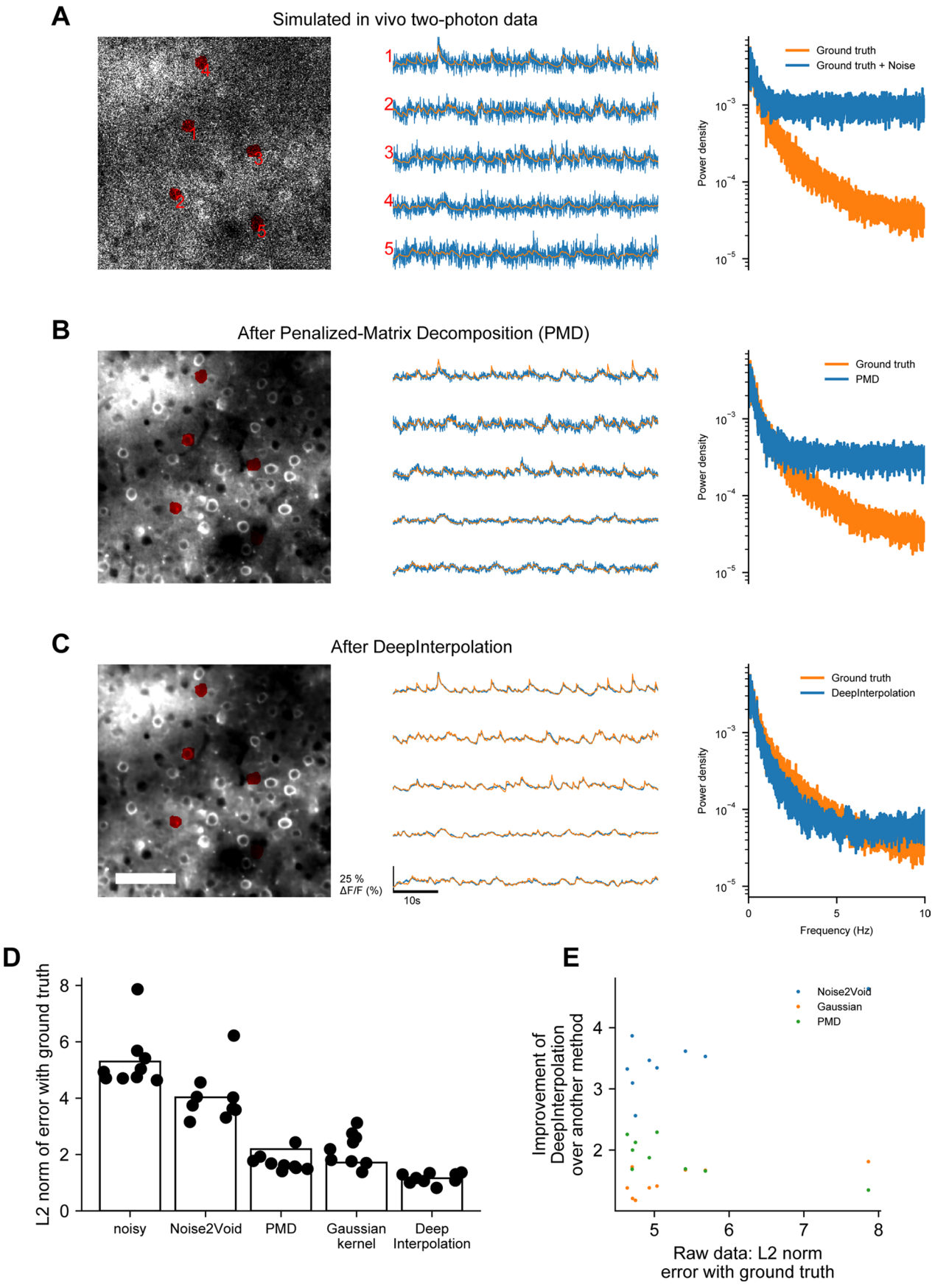
Extended data is available for this paper at https://doi.org/10.1038/s41592-021-01285-2.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41592-021-01285-2.

Correspondence and requests for materials should be addressed to Jérôme Lecoq.

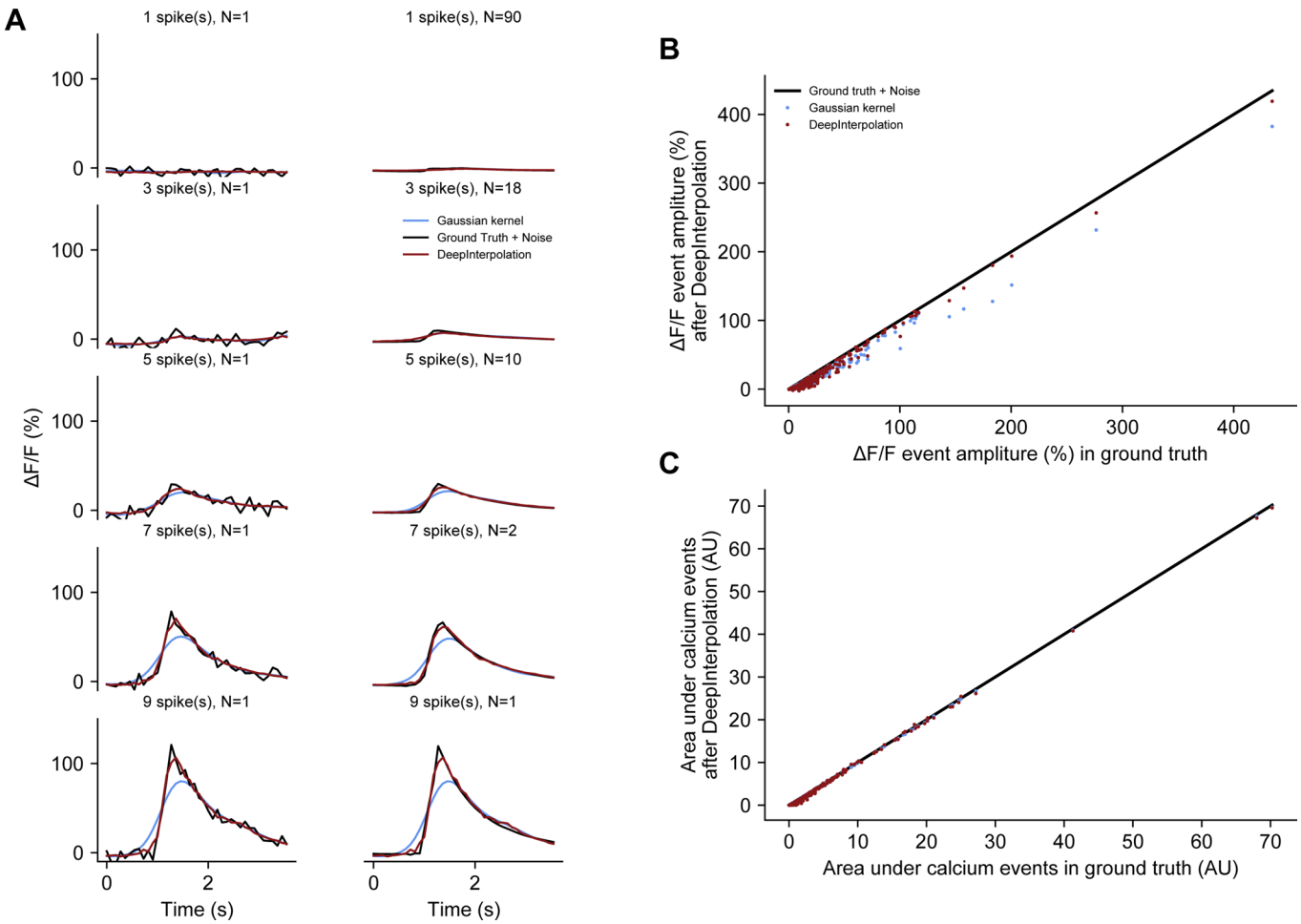
Peer review information *Nature Methods* thanks Adam Packer and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Nina Vogt was the primary editor on this article and managed its editorial process and peer review in collaboration with the rest of the editorial team.

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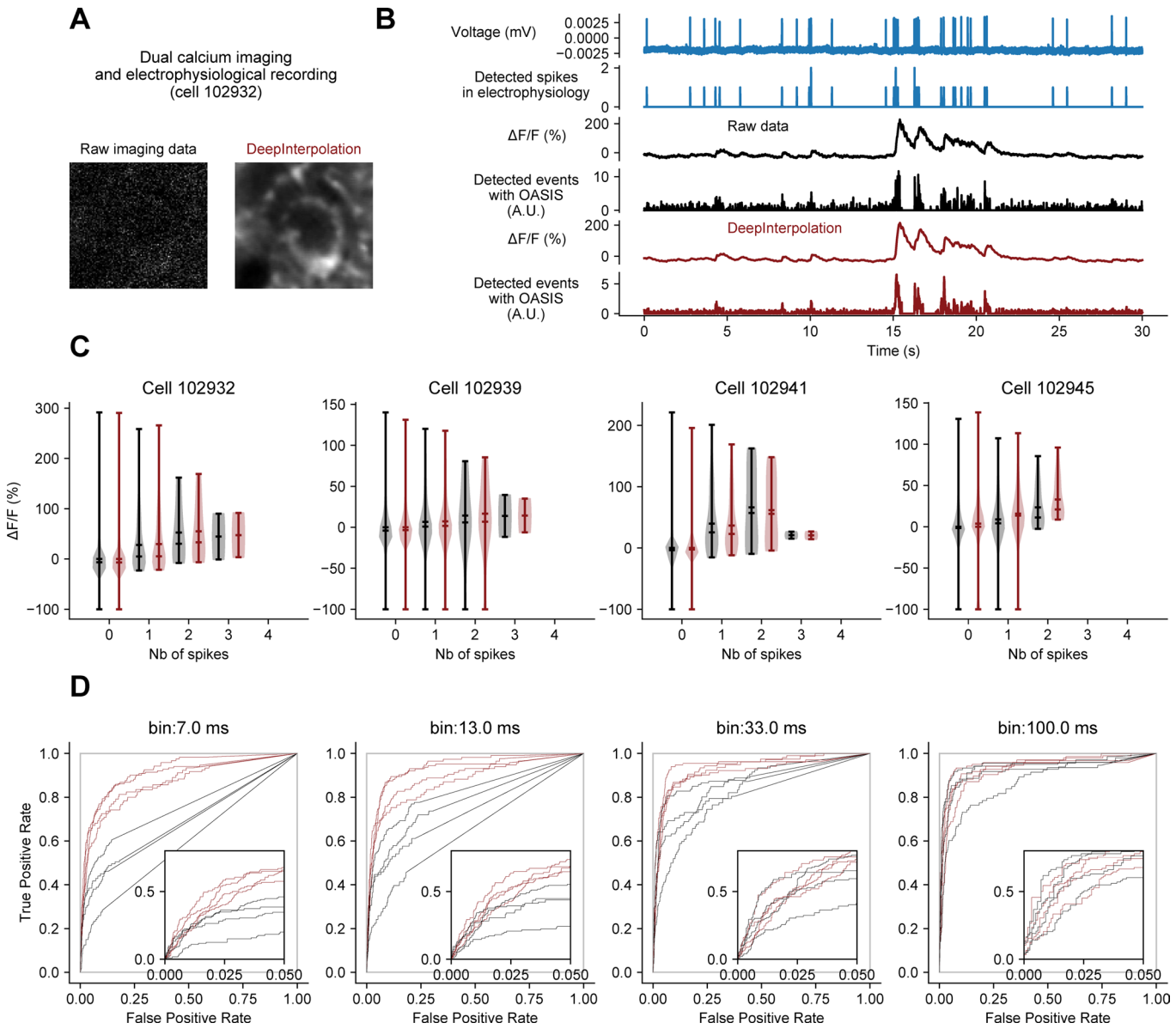


Extended Data Fig. 1 | See next page for caption.

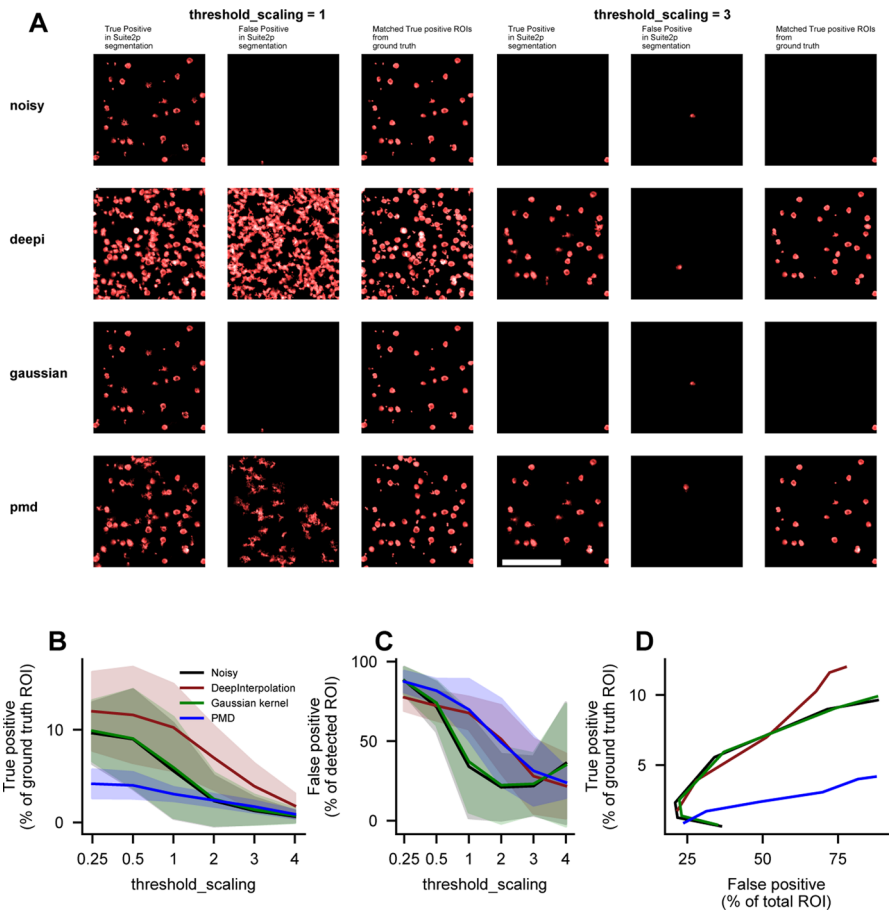
Extended Data Fig. 1 | Comparison of DeepInterpolation, Noise2Void[6], a Gaussian kernel, Penalized Matrix Decomposition (PMD)[2] on simulated calcium movies with ground truth. (**a**) A simulated two photon calcium movie that was generated using an *in silico* Neural Anatomy and Optical Microscopy (NAOMi) simulation[12]. **Left**: A few ROIs were manually drawn to extract the associated traces (**middle**) of the simulated movie with (blue trace) and without sources of noise (ground truth, orange). **Right**: The corresponding Fourier transform of all traces was calculated for each trace and averaged across ROIs. (**b**) The same movie and traces after PMD denoising. (**c**) The same movie and traces after DeepInterpolation. Scale bar is 100 μm. (**d**) Quantification of average mean squared error (L2) between noisy traces and ground truth, as well as between Noise2Void, PMD, Gaussian kernel (with $\sigma = 3$) and DeepInterpolation with ground truth. (**e**) quantification of the improvement in L2 norm against the original noisy error.



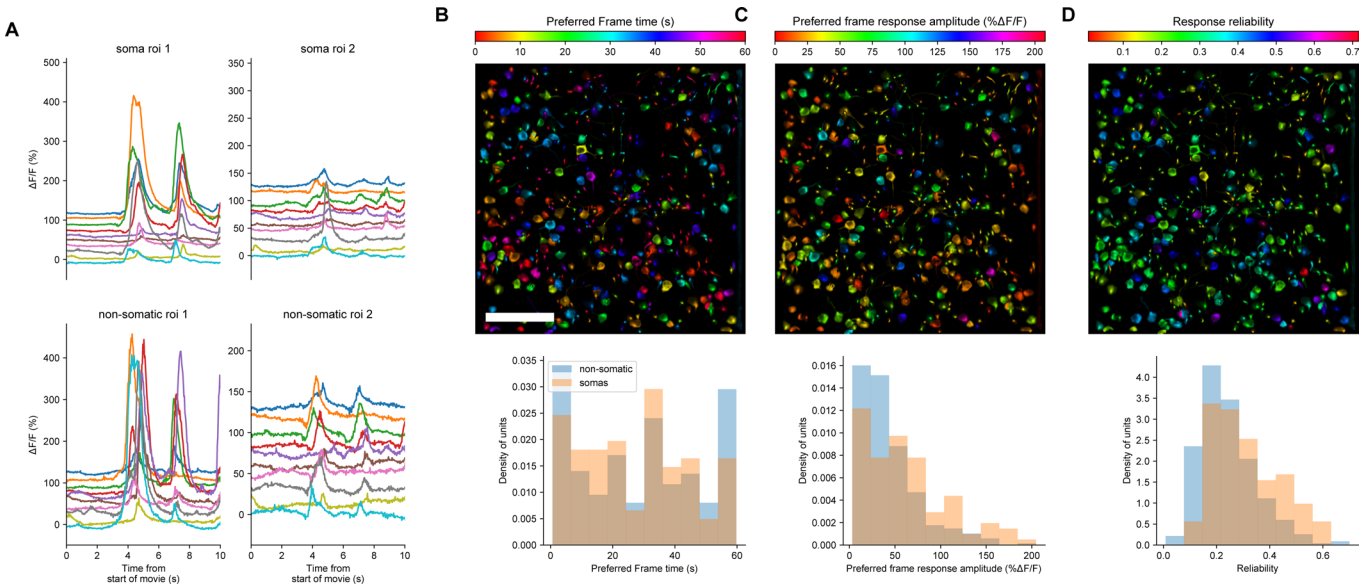
Extended Data Fig. 2 | Analysis of spike-evoked calcium responses in simulated calcium movies with ground truth. (**a**) For the simulated movie in Supplementary Fig. 2, spike-triggered traces were generated at different levels of activity. Spikes were counted within 272 ms windows and used to sort calcium events. Left column corresponds to a single trial associated with one spiking event. Right column is the average of all associated trials. Traces were extracted from (**b**). For all neuronal compartments created in the field of view, the peak calcium spike amplitude was measured in ground truth data as well as in movies that went through DeepInterpolation and Gaussian kernel. Red line is the identity line. (**c**) Same plot as (**b**) except that the area under all calcium events was measured.



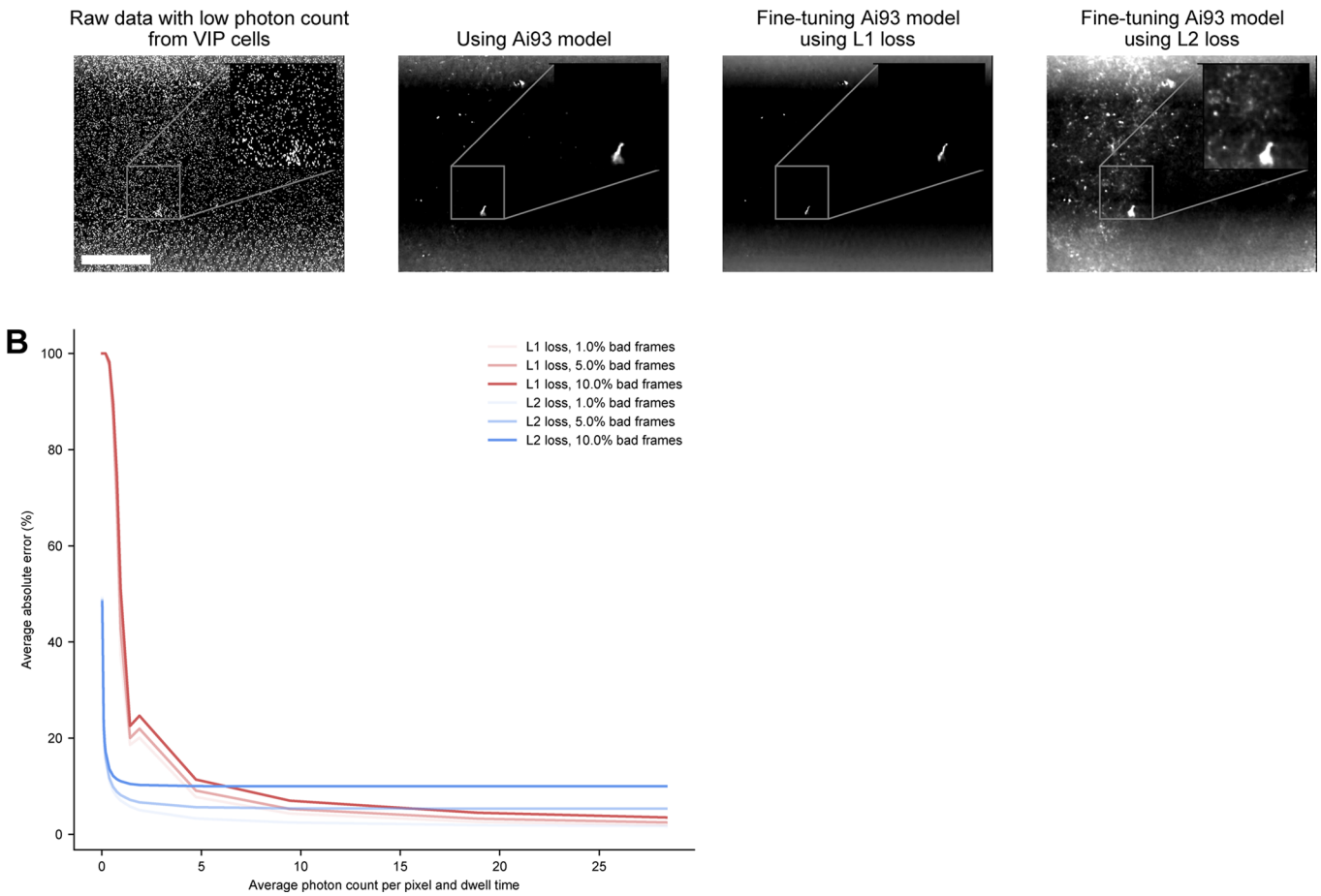
Extended Data Fig. 3 | Quantification of DeepInterpolation impact on spike detection performance using high-quality simultaneous optical and cell-attached electrophysiological recording. (**a**) Example high-magnification single two-photon frame acquired during a dual calcium imaging and cell-attached cellular recording in vivo in a Camk2a-tTA; tetO-GCaMP6s transgenic mouse. Cell id throughout the figure refers to the available public dataset cell ids[14] (**b**) Exemplar simultaneous voltage cellular recording, detected Action Potential (AP), original fluorescence recording, denoised trace with DeepInterpolation and detected events using fast non-negative deconvolution[39] based either on the original or the denoised data. (**c**) Violin distribution plots of ΔF/F amplitude associated with number of recorded action potentials in 100 ms bins. Raw traces (black) and traces after DeepInterpolation (red) are shown for 4 different recorded cells. (**d**) Receiver operating characteristic (ROC) curves showing the detection probability for true APs against probability of false positives as detection threshold is changed, for one or more spikes. ROC curves were calculated for 4 different temporal bin sizes. False positives were calculated from time windows with no spikes. Each ROC curve represents a different cell id. Red curves are obtained after DeepInterpolation. Inset plots zoom on lower false positive rates.



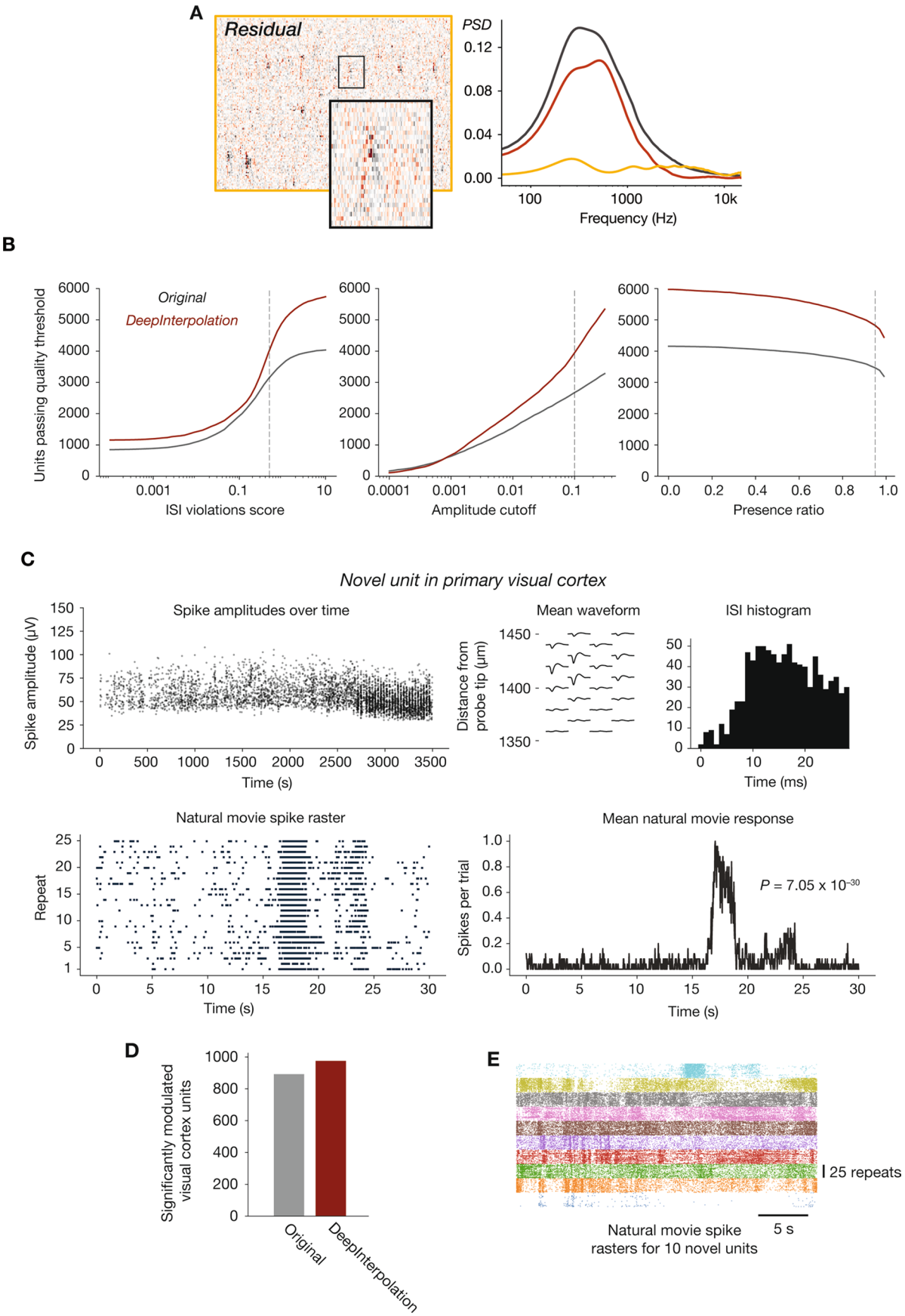
Extended Data Fig. 4 | Quantification of DeepInterpolation impact on segmentation performance of Suite2p with synthetic data from NAOMi. (**a**) The noisy raw movie (first row) was pre-processed with either a fine tuned DeepInterpolation model (second row), a gaussian kernel (third row), or using Penalized Matric Factorization (PMD) (fourth row). The first column shows the ROIs or cell filters identified by Suite2p that were matched with corresponding ideal ROIs in the ground truth synthetic data (cross-correlation > 0.7). The second column shows ROIs that could not be matched (cross-correlation < 0.7) with any ideal ROI in the synthetic data. The third column shows the ideal ROIs projection in the synthetic data that were matched with a segmented filter (that is not the ROI profile that were found using suite2p but the expected ROI from the simulation). Two sets of segmentation are displayed for two values of threshold_scaling, a key parameter in suite2p controlling the detection sensitivity. Scale bar is 100 µm. (**b**) True positive rate, that is the proportion of ROIs in the simulation that were found by suite2p. threshold_scaling was varied from 0.25 to 4 (default is 1) for every datasets. N = 4 simulated movies. Shaded area represents the standard deviation across 4 simulations. (**c**) False positive, that is the proportion of found ROIs that were not found in the simulated set of ideal ROIs (cross-correlation <0.7). N = 4 simulated movies. Shaded area represents the standard deviation across 4 simulations. (**d**) True positive rate against false positive rate for each pre-processing method.



Extended Data Fig. 5 | Natural movie responses extracted from somatic and non-somatic compartments after DeepInterpolation. (**a**) Example calcium responses to 10 repeats of a natural movie for 4 examples ROIs, both somatic and non-somatic (**b,c,d**) **Top:** Analysis of the sensory response to 10 repeats of a natural movie visual stimulus. Each ROI detected after DeepInterpolation was colored by the preferred movie frame time (**b**), the amplitude of the calcium response at the preferred frame (**c**) and the response reliability (**d**). Scale bar is 100 μm. (**b**) **Bottom:** Distribution of preferred frame of the natural movie for both somatic and non-somatic ROIs in the same field of view. (**c**) **Bottom:** Same as (B) for response amplitude at the preferred time. (**d**) **Bottom**: Reliability of all ROIs response to the natural movie. Reliability was computed by averaging all pairwise cross-correlation between each individual trial.

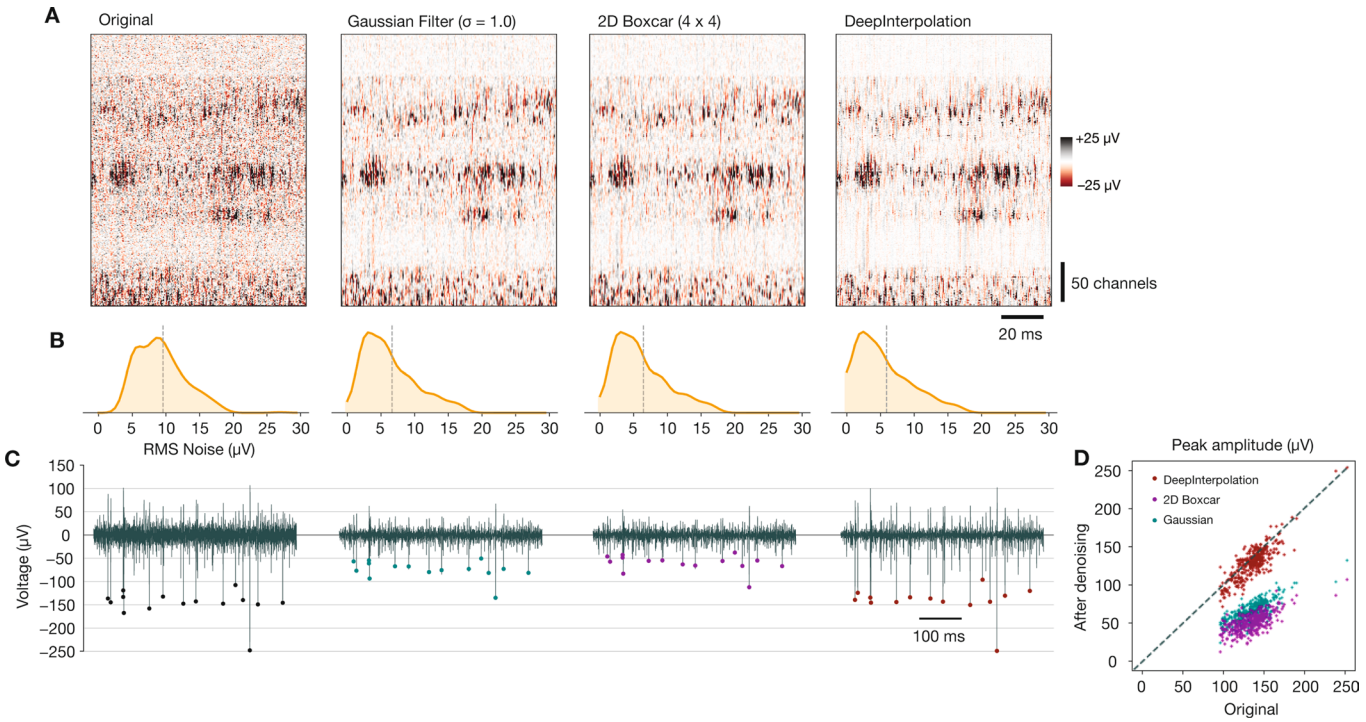


Extended Data Fig. 6 | Fine tuning DeepInterpolation models with an L2 loss yields better reconstruction when photon count is very low. (**a**) Comparison of an example image in a two-photon movie showing VIP cells expressed GCaMP6s in Raw data, after DeepInterpolation with a broadly trained model (Ai93), after DeepInterpolation with a model trained on VIP data with an L1 loss and after DeepInterpolation with a model trained on VIP data with an L2 loss. Scale bar is 100 μm. (**b**) Simulation of reconstruction error when using either the L1 loss or the L2 loss for various levels of photon count and corrupted frames.

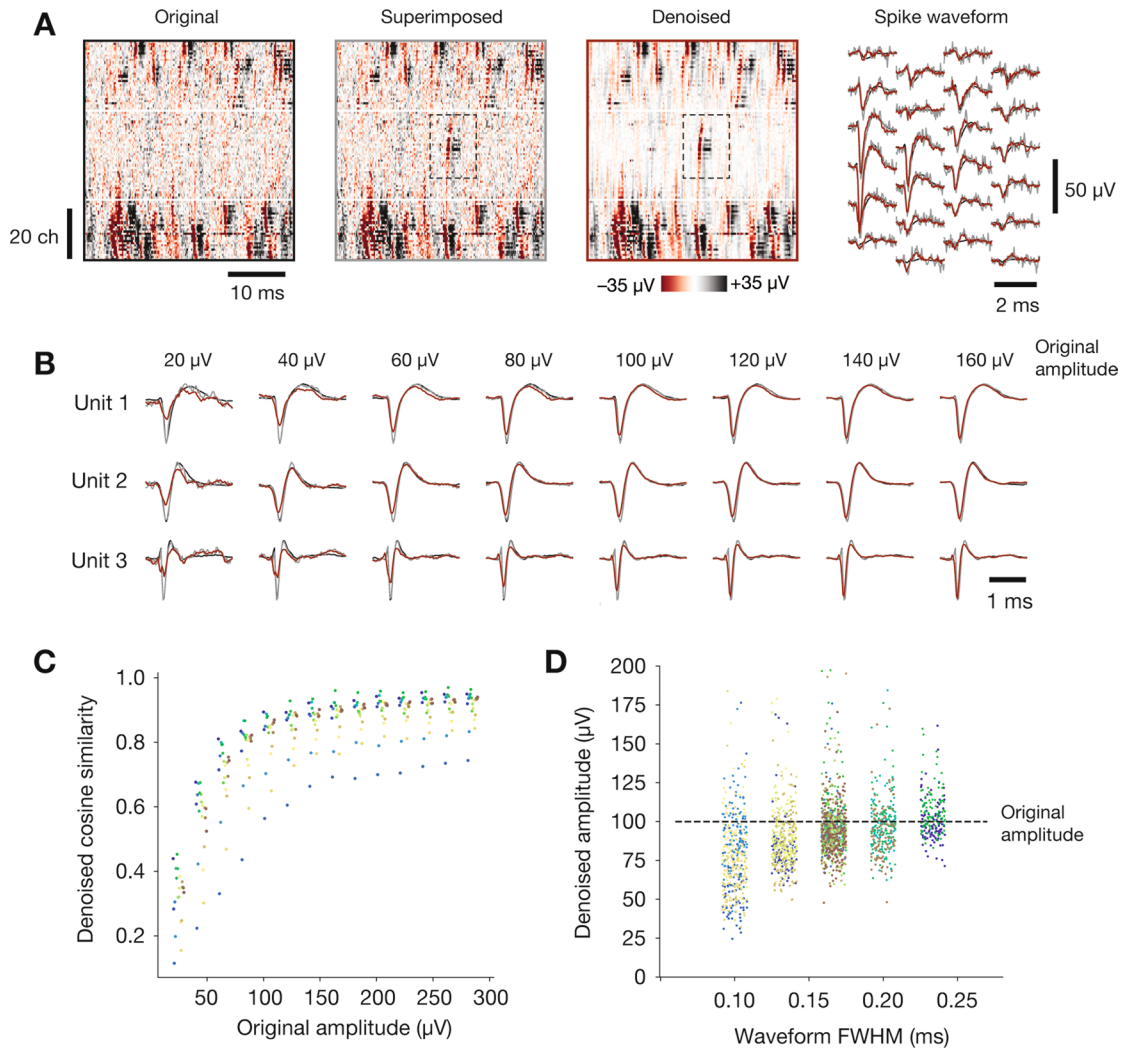


Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Extended data for Neuropixels recordings. (**a** *Left:* Heatmap of the residual (denoised subtracted from the original data), for the same data shown in Fig. 3b. *Right:* Power spectra for the original (gray), denoised (red), and residual (yellow) data. (**b**) Number of units passing a range of quality control thresholds, for three different unit quality metrics. QC thresholds used in this work are indicated with dashed lines. (**c**) Physiological plots for one example unit from V1 that was only detectable after denoising (<20% of spike times were included in the original spike sorting results). (*P*-value calculated using the Kolmogorov–Smirnov test between the distribution of spikes per trial for the original and shuffled response). (**d**) Number of units with substantial response modulation to a repeated natural movie stimulus, for all cortical units detected before and after denoising. (**e**) Raster plots for 10 high-reliability exemplar units detected only after denoising, aligned to the 30 s natural movie clip. Each color represents a different unit.



Extended Data Fig. 8 | Comparing DeepInterpolation with traditional denoising methods. (**a**) Heatmap of 100 ms of Neuropixels data before and after applying three denoising methods: Gaussian filter, 2D Boxcar, and DeepInterpolation. (**b**) Distribution of RMS noise levels across channels shown in (A), computed over the same 10 s interval. The parameters for the Gaussian and Boxcar filters were chosen such that the resulting RMS noise distributions roughly matched that seen with DeepInterpolation. (**c**) Example voltage time series for one channel, before and after applying the three denoising methods shown in (A). (**d**) Relationship between original peak amplitude and denoised peak amplitude for all spikes in a 10 s window, from the same example channel shown in (**c**).



Extended Data Fig. 9 | Results of simulated ground truth electrophysiology experiments. (**a**) Right: Heatmap of original data (black outline), data with superimposed spike waveforms (gray outline), and superimposed data after denoising (red outline). Left: Spike waveform outlined in heatmaps, for three different conditions. (**b**) Mean peak waveforms for each of three units superimposed on the original data at 8 different amplitudes (N = 100 spikes per unit per amplitude). (**c**) Mean cosine similarity between individual denoised waveforms and the original superimposed waveform. Colors represent different units. (**d**) Relationship between denoised amplitude and waveform full-width at half max (FWHM), for all 100 µV waveforms. Colors represent different units.

nature research

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No data was collected as part of this study and we therefore did not used associated software for data collection. We used data previously published and made available to the community.
Data analysis	We released 3 types of code/data with this publication. (1) The DeepInterpolation package for training and inference of DeepInterpolation networks https://github.com/AllenInstitute/deepinterpolation. A toy dataset was provided in the repository to introduce everyone to use the package. (2) A repository to re-create all figures created for the manuscript using python code : https://github.com/AllenInstitute/deepinterpolation_paper. All intermediary and necessary datasets are provided as downloadable links with the repository. The electrophysiology figures can be re-generated in sub-panels. (3) pre-trained networks for inference with Tensorflow are available through the github reposity at: https://github.com/AllenInstitute/deepinterpolation. We used Tensorflow 1.9-2.2 and Python 3.6-3.8 throughout this work. PMD was ran as part of a docker environment (https://hub.docker.com/r/paninski/trefide). NAOMI v0.1 was used in Matlab 2020a .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

- We provided all intermediary raw data as downloadable links with the repository to regenerate all figures https://github.com/AllenInstitute/deepinterpolation_paper
- The two-photon imaging and Neuropixels raw data can be downloaded from the following S3 bucket: arn:aws:s3:::allen-brain-observatory
Two-photon imaging files are accessed according to Experiment ID, using the following path:
visual-coding-2p/ophys_movies/ophys_experiment_<Experiment ID>.h5
We used a random subset of 1144 experiments for training the denoising network for Ai93, and 397 experiments for training the denoising network for Ai148. The list of used experiments IDs is available in json files (in deepinterpolation/examples/json_data/) on the DeepInterpolation GitHub repository. The majority of these experiment IDs are available on the S3 bucket. Some experimental data has not been released to S3 at the time of publication.
Neuropixels raw data is accessed by Experiment ID and Probe ID, using the following paths:
visual-coding-neuropixels/raw-data/<Experiment ID>/<Probe ID>/spike_band.dat
The dat files have the median subtraction post-processing applied, but do not include an offline highpass filter. Prior to DeepInterpolation, we extracted 3600 seconds of data from each of the recordings listed in Supp. Table 1.
To train the fMRI denoiser, we used datasets that can be downloaded from an S3 bucket:
arn:aws:s3:::openneuro.org/ds001246
We trained our denoiser on all "ses-perceptionTraining" sessions, across 5 subjects (3 perception training sessions, 10 runs each).
- Pre-trained DeepInterpolation models were made available through the Deepinterpolation package repository https://github.com/AllenInstitute/deepinterpolation.
- There is no restriction on data availability. All datasets are already public.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	no sample-size calculation was performed. We evaluated the stability of validation loss during training through its standard deviation across training runs. This allowed to determine the size of validation datasets. The amount of training data used was determined using the convergence of the validation loss across training samples.
Data exclusions	No data was excluded.
Replication	All training performance was measured on held out datasets during validation ensuring generalization of our performance. We provided analysis of denoising performance on datasets acquired in new recording condition, ensuring duplication by external laboratories. We held out datasets for inference to measure the validation performance of the DeepInterpolation models.
Randomization	Samples were randomized during training of DeepInterpolation networks. Sample size is provided along with the exact experiment ID and frame numbers used in json files on the github repository.
Blinding	Blinding was applied using traditional machine learning approaches where a subset of the data was held out to measure performance. No data was acquired during this study and therefore data collection procedures did not influence our results.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Magnetic resonance imaging

Experimental design

Design type	Data was collected in prior publication and is appropriate cited.
Design specifications	Data was collected in prior publication and is appropriate cited.
Behavioral performance measures	Data was collected in prior publication and is appropriate cited.

Acquisition

Imaging type(s)	Data was collected in prior publication and is appropriate cited.
Field strength	Data was collected in prior publication and is appropriate cited.
Sequence & imaging parameters	Data was collected in prior publication and is appropriate cited.
Area of acquisition	Data was collected in prior publication and is appropriate cited.

Diffusion MRI ☐ Used ☐ Not used

Preprocessing

Preprocessing software	No preprocessing. We used raw data provided by a previous publication.
Normalization	Normalization was using during training but re-applied after inference. An estimate of mean and standard deviation was calculated to center the data and is described in details in the method section.
Normalization template	N/A
Noise and artifact removal	Noise removal is described throughout the publication as this is the main conceptual advance of our work.
Volume censoring	N/A

Statistical modeling & inference

Model type and settings	N/A
Effect(s) tested	N/A

Specify type of analysis: ☒ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference (See Eklund et al. 2016)	N/A
Correction	N/A

Models & analysis

n/a	Involved in the study
☒	☐ Functional and/or effective connectivity
☒	☐ Graph analysis
☒	☐ Multivariate modeling or predictive analysis